Functions, domains and ranges

Inputs, outputs and restrictions

A function assigns exactly one output to each allowed input. Its *domain* is the set of allowed inputs; its *range* is the set of outputs actually attained. For a function of two variables, an input is a point (x, y).
For real-valued formulas, use these restrictions together:

$$\sqrt{A} \text{ requires } A \geq 0, \qquad \frac{1}{A} \text{ requires } A \neq 0, \qquad \ln A \text{ requires } A > 0.$$

An inverse reverses a one-to-one function: $p^{-1}(p(t)) = t$ on its domain. A composition $(p \circ q)(t) = p(q(t))$ is defined only when t is allowed for q *and* $q(t)$ is allowed for p.

Basic graph	Domain	Range
$y = x^2$	$\mathbb{R}$	$[0, \infty)$
$y = x^3$	$\mathbb{R}$	$\mathbb{R}$
$y = 1/x$	$\mathbb{R} \setminus \{0\}$	$\mathbb{R} \setminus \{0\}$
$y = e^x$	$\mathbb{R}$	$(0, \infty)$
$y = \ln x$	$(0, \infty)$	$\mathbb{R}$
$y = \sin x$	$\mathbb{R}$	$[-1, 1]$

Example 1. An inverse does not remove domain restrictions

Let

$$p(t) = \ln(t - 1), \qquad q(t) = \sqrt{t + 2}.$$

Describe their basic graphs, domains and ranges. Find their inverses, and find both $p \circ q$ and $q \circ p$ with their domains.

Solution.

The two original functions

For p, the logarithm requires $t - 1 > 0$, so $t > 1$. Its graph is the logarithm graph shifted one unit to the right: it passes through $(2, 0)$, increases, and has vertical asymptote $t = 1$. Every real output occurs. Thus

$$\operatorname{Dom} p = (1, \infty), \qquad \operatorname{Range} p = \mathbb{R}.$$

For q, $t + 2 \geq 0$, so $t \geq -2$. Its graph starts at $(-2, 0)$, passes through $(-1, 1)$ and increases. A principal square root is nonnegative:

$$\operatorname{Dom} q = [-2, \infty), \qquad \operatorname{Range} q = [0, \infty).$$

Invert by solving for the original input

From $s = \ln(t - 1)$, exponentiation gives $t - 1 = e^s$. Therefore

$$p^{-1}(s) = 1 + e^s, \qquad s \in \mathbb{R}.$$

From $s = \sqrt{t + 2}$, we have $s \geq 0$ and $s^2 = t + 2$. Therefore

$$q^{-1}(s) = s^2 - 2, \qquad s \geq 0.$$

The restriction $s \geq 0$ is essential: the original square root never produces a negative output.

Compose, then check every restriction

For the first composition,

$$(p \circ q)(t) = \ln(\sqrt{t + 2} - 1).$$

The logarithm requires $\sqrt{t + 2} > 1$, which is equivalent to $t > -1$. This also ensures that the square root is defined.

For the other order,

$$(q \circ p)(t) = \sqrt{\ln(t-1)+2}.$$

We need $t > 1$ and $\ln(t-1)+2 \geq 0$. The latter gives $t - 1 \geq e^{-2}$, so

$$\boxed{\begin{aligned} &\mathrm{Dom}(p \circ q) = (-1, \infty), \\ &\mathrm{Dom}(q \circ p) = [1 + e^{-2}, \infty). \end{aligned}}$$

The second endpoint is included because a square root may equal zero; the logarithm's argument there is $e^{-2} > 0$.

Interior, boundary and boundedness

An interior point has a small open disk around it lying entirely in the set. A boundary point has points of the set and of its complement arbitrarily close to it. A set is *open* if all its points are interior points, and *closed* if it contains all its boundary points.
A planar set is *bounded* if it fits inside a sufficiently large disk. Open/closed and bounded/unbounded are separate questions. A set can also be neither open nor closed.

Example 2. Two square roots give different inequalities

Find the domain and range of

$$F(x, y) = \frac{\sqrt{x-y}}{\sqrt{2-x-y}}.$$

Describe its boundary and classify its domain as open/closed and bounded/unbounded.

Solution.

Build the domain from both square roots

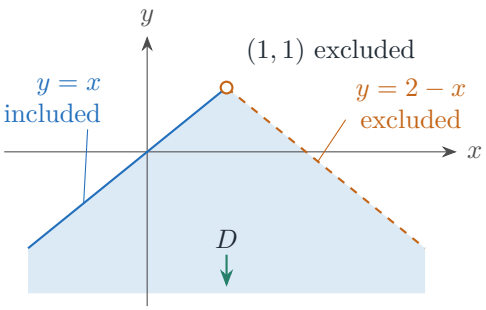
The numerator requires $x - y \geq 0$. The denominator's square root must be strictly positive, so $2 - x - y > 0$. Hence

$$\boxed{D = \{(x, y) : y \leq x, \quad y < 2 - x\}.}$$

The two lines meet at $(1, 1)$. The boundary consists of

$$y = x, \ x \leq 1, \qquad \text{and} \qquad y = 2 - x, \ x \geq 1.$$

The first ray is included for $x < 1$. The second ray and the meeting point $(1, 1)$ are excluded.



The region continues downwards.

Classify the domain

The point $(0, 0)$ belongs to D but lies on its boundary, so D is not open. Points $(1, 1 - 1/n)$ belong to D and approach the excluded point $(1, 1)$, so D is not closed.

For every $t \geq 0$, $(t, -t) \in D$, and its distance from the origin grows without bound. Thus D is unbounded.

$$\boxed{D \text{ is neither open nor closed, and is unbounded.}}$$

Show which outputs are actually attained

The quotient is nonnegative. Along the allowed line $y = -x$, with $x \geq 0$,

$$F(x, -x) = \frac{\sqrt{2x}}{\sqrt{2}} = \sqrt{x}.$$

For any desired output $s \geq 0$, the point $(s^2, -s^2)$ therefore gives $F = s$. Consequently

$$\boxed{\text{Range } F = [0, \infty).}$$

This last construction proves that every value in the proposed range occurs.

Example 3. A bounded domain need not give bounded outputs

(a) Find the range of $f(x, y) = \tan x$ on $D_1 = (-\pi/2, \pi/2) \times (0, 1)$.

(b) Find the range of $g(x, y) = xe^{-y}$ on $D_2 = [-1, 1] \times [0, 2]$.

(c) Classify $U = \{(x, y) : y > e^x\}$ and $C = \{(x, y) : y \geq e^x\}$ as open/closed and bounded/unbounded.

Solution.

(a) Bounded inputs, unbounded output

The rectangle D_1 is open and bounded. Nevertheless, $\tan x$ approaches $+\infty$ as $x \to (\pi/2)^-$ and $-\infty$ as $x \to (-\pi/2)^+$. For any $s \in \mathbb{R}$, choose $x = \arctan s$ and $y = 1/2$. This point belongs to D_1 and has output s. Thus

$$\boxed{f(D_1) = \mathbb{R}.}$$

(b) A continuous function on a closed bounded domain

On D_2, $|x| \leq 1$ and $0 < e^{-y} \leq 1$. Therefore $|xe^{-y}| \leq 1$. Every $s \in [-1, 1]$ is attained at $(x, y) = (s, 0)$, so

$$\boxed{g(D_2) = [-1, 1].}$$

The minimum -1 and maximum 1 are attained at $(-1, 0)$ and $(1, 0)$. This illustrates the extreme-value theorem: a continuous real-valued function on a nonempty closed bounded region attains a finite maximum and minimum.

(c) The same boundary, different inclusion

Both sets have boundary $y = e^x$. The strict inequality defines the open region above that curve; the non-strict inequality includes the entire boundary, giving a closed region. Small changes preserve a strict positive value of $y - e^x$ because this function is continuous.

Both sets are unbounded: $(0, n)$ belongs to both for every integer $n \geq 2$. Thus U is open and unbounded, while C is closed and unbounded.

Check. Boundedness describes where inputs lie. It does not by itself control outputs. Continuity together with a closed bounded domain supplies the stronger conclusion used in part (b).

Graphs, traces and level curves

Three related geometric objects

The graph of $z = f(x, y)$ consists of points $(x, y, f(x, y))$ in three-dimensional space. A trace is obtained by fixing one coordinate, such as $x = a$. A level curve $f(x, y) = c$ lies in the input plane: all its points give the same output c.
A contour map shows several level curves. When contour values have equal increments, their spacing helps show how quickly the output changes in a specified direction.

Example 4. V-shaped traces and diamond-shaped contours

For

$$z = f(x, y) = |x| + 2|y|,$$

find the traces in $x = 0$ and $y = 0$. Describe the level sets for $c = -1, 0, 2, 4$, and draw the nonempty ones. Find two distinct points on the $c = 4$ contour. Compare the output changes for equal displacements from the origin along the positive x- and y-axes.

Solution.

Fix a coordinate to obtain a trace

In the vertical plane $x = 0$,

$$z = 2|y|,$$

a V-shaped graph with vertex at the origin. In the plane $y = 0$,

$$z = |x|,$$

which is another V-shaped graph. These traces lie on the three-dimensional graph; they are not contours in the xy-plane.

Fix an output to obtain a level curve

Setting $f = c$ gives

$$|x| + 2|y| = c.$$

The left side is nonnegative, so the level $c = -1$ is empty. When $c = 0$, both terms must vanish; this level consists only of $(0, 0)$.

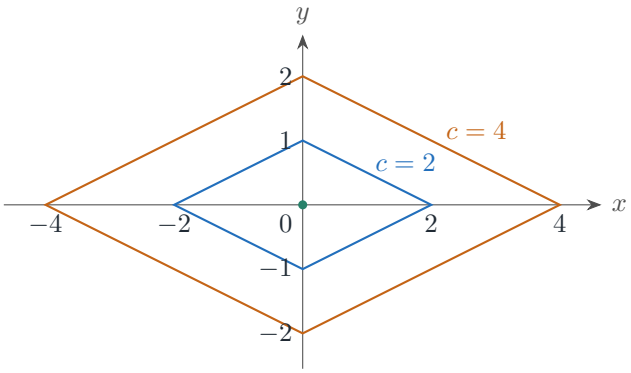
For $c > 0$, first work in the first quadrant. There $|x| = x$ and $|y| = y$, so the contour is the line segment

$$x + 2y = c, \qquad x \geq 0, \quad y \geq 0,$$

joining $(c, 0)$ to $(0, c/2)$. Reflection in both coordinate axes gives a diamond with vertices

$$(c, 0), \quad (0, c/2), \quad (-c, 0), \quad (0, -c/2).$$

Thus the $c = 2$ diamond has vertices $(\pm 2, 0), (0, \pm 1)$, while the $c = 4$ diamond has vertices $(\pm 4, 0), (0, \pm 2)$.



$c = 0$: the origin only

For example, $(4,0)$ and $(0,2)$ both give output 4. These are two input points on the $c=4$ contour. The corresponding points on the surface are $(4,0,4)$ and $(0,2,4)$.

Interpret the contour spacing

For the same displacement $d>0$ from the origin along the positive axes,

$$f(d,0)-f(0,0)=d, \qquad f(0,d)-f(0,0)=2d.$$

The output increases twice as much along the positive y-axis. Consistently, moving from level 2 to level 4 takes 2 units along the positive x-axis, but only 1 unit along the positive y-axis.

Limits and continuity

What a two-variable limit requires

The statement $\lim_{(x,y)\to(a,b)} f(x,y)=L$ means that the output approaches L along every allowed approach to (a,b), not just along the coordinate axes. Two different path limits prove nonexistence. Agreement along a few paths does not prove existence.
Continuous algebraic combinations allow direct substitution where defined; for a quotient, the limiting denominator must be nonzero. To prove a zero limit, a bound $|f(x,y)|\le B(x,y)$ with $B\to 0$ is often more useful than testing paths.

Example 5. A direct limit and an estimate valid in every direction

Evaluate

$$\text{(a)} \quad \lim_{(x,y)\to(0,1)} \frac{e^{xy}+x^2}{2+y}, \qquad \text{(b)} \quad \lim_{(x,y)\to(0,0)} \frac{x^2\sin y}{x^2+y^2}.$$

Solution. For (a), the denominator approaches $3\neq 0$. The numerator approaches $e^0+0=1$, so direct substitution gives

$$\boxed{\frac{1}{3}}.$$

For (b), direct substitution gives $0/0$ and is not a limit rule. Use $x=r\cos\theta$, $y=r\sin\theta$, where $r>0$ and $r\to 0$ as $(x,y)\to(0,0)$. Then

$$\frac{x^2\sin y}{x^2+y^2}=\cos^2\theta\,\sin(r\sin\theta).$$

The elementary inequality $|\sin u|\le|u|$ gives

$$\left|\cos^2\theta\,\sin(r\sin\theta)\right|\le|\cos^2\theta|\,r|\sin\theta|\le r.$$

The final bound tends to zero and is independent of θ. Therefore it covers directions that change as the point approaches the origin:

$$\boxed{\lim_{(x,y)\to(0,0)} \frac{x^2\sin y}{x^2+y^2}=0.}$$

Check. A Cartesian estimate also works: $|x^2\sin y/(x^2+y^2)|\le|\sin y|\le|y|\to 0$.

Example 6. Even all straight-line tests can miss the problem

Determine whether

$$\lim_{(x,y)\to(0,0)} \frac{x^2(y-x)}{x^4+(y-x)^2}$$

exists. Compare straight lines through the origin with a suitable curved path.

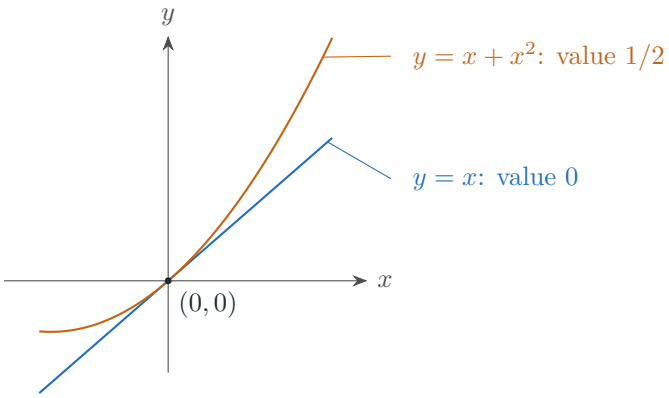
Solution. The denominator vanishes only at the origin. On the line $y = mx$, for $m \neq 1$ and $x \neq 0$, the expression becomes

$$\frac{x^2(m-1)x}{x^4+(m-1)^2x^2} = \frac{(m-1)x}{x^2+(m-1)^2} \longrightarrow 0.$$

On $y = x$ it is zero, and on the vertical line $x = 0$ it is also zero. Thus every straight-line test gives zero.

The denominator suggests balancing x^4 with $(y-x)^2$. Choose $y - x = x^2$, that is, $y = x + x^2$. Along this curve, for $x \neq 0$,

$$\frac{x^2(y-x)}{x^4+(y-x)^2} = \frac{x^4}{x^4+x^4} = \frac{1}{2}.$$



Both paths approach the origin but give different limits. Hence

The two-variable limit does not exist.

The curved path detects behaviour that every straight line misses.

Continuity at a point

A function is continuous at (a, b) when it is defined there and

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b).$$

When a formula is replaced by a special value at one point, that value must equal the surrounding limit.

Example 7. Choose the value that makes the function continuous

Let

$$G(x,y) = \begin{cases} \dfrac{e^{x^2+y^2}-1}{x^2+y^2}, & (x,y) \neq (0,0), \\ c, & (x,y) = (0,0). \end{cases}$$

Choose c so that G is continuous everywhere.

Solution. Away from the origin, the denominator is positive and the formula is a quotient of continuous functions. Only the origin needs attention.

Put $u = x^2 + y^2$. Then $u \to 0^+$ as $(x, y) \to (0, 0)$. The familiar one-variable derivative of e^u at $u = 0$ gives

$$\lim_{u \to 0} \frac{e^u - e^0}{u - 0} = 1.$$

Consequently

$$\lim_{(x,y) \to (0,0)} G(x, y) = 1.$$

Continuity requires the point value to equal this limit, so

$$\boxed{c = 1}.$$

With this choice, both the ordinary formula points and the origin are points of continuity.

Partial derivatives and local sensitivity

Hold the other independent input fixed

For a function $f(x, y)$, the first partial derivatives at (a, b) are

$$f_x(a, b) = \lim_{h \to 0} \frac{f(a + h, b) - f(a, b)}{h}, \qquad f_y(a, b) = \lim_{k \to 0} \frac{f(a, b + k) - f(a, b)}{k},$$

provided the limits exist. In f_x, hold y fixed; in f_y, hold x fixed. Ordinary product, quotient and chain rules then apply.
On the graph $z = f(x, y)$, $f_x(a, b)$ is the tangent slope of the trace with $y = b$, and $f_y(a, b)$ is the tangent slope of the trace with $x = a$.

Example 8. Differentiate one input at a time

For

$$f(x, y) = x \ln(1 + y^2) + \frac{y}{1 + x},$$

find f_x, f_y, state where the formulas apply, and interpret their values at $(1, 1)$ as trace slopes.

Solution. Since $1 + y^2 > 0$ for all real y, only the denominator restricts the domain: $x \neq -1$.

For f_x, $\ln(1 + y^2)$ and y are constants:

$$\begin{aligned} f_x &= \frac{\partial}{\partial x}[x \ln(1 + y^2)] + y \frac{\partial}{\partial x}(1 + x)^{-1} \\ &= \ln(1 + y^2) - \frac{y}{(1 + x)^2}. \end{aligned}$$

For f_y, x and $1 + x$ are constants. The logarithm needs the chain rule:

$$\begin{aligned} f_y &= x \frac{1}{1 + y^2}(2y) + \frac{1}{1 + x} \\ &= \frac{2xy}{1 + y^2} + \frac{1}{1 + x}. \end{aligned}$$

Both formulas are valid when $x \neq -1$. At $(1, 1)$,

$$\boxed{f_x(1, 1) = \ln 2 - \frac{1}{4}, \qquad f_y(1, 1) = \frac{3}{2}.}$$

The point on the graph is $(1, 1, \ln 2 + 1/2)$. The first value is the tangent slope in the vertical plane $y = 1$; the second is the tangent slope in the vertical plane $x = 1$.

Example 9. Sensitivity of heat loss to a panel's dimensions

A rectangular panel has side lengths a, b in metres. A simplified model for its heat loss, in watts, is

$$Q(a,b) = 5ab + 2(a+b), \qquad a, b > 0.$$

Find Q_a, Q_b at $(a,b) = (2,1)$. Which of two equally small increases in side length has the larger effect? Estimate the change when b increases by 0.02 m while a stays fixed.

Solution. Differentiate while holding the other side length fixed:

$$Q_a = 5b + 2, \qquad Q_b = 5a + 2.$$

At $(2,1)$,

$$\boxed{Q_a = 7 \text{ W/m}, \qquad Q_b = 12 \text{ W/m}.}$$

Both derivatives are positive, so increasing either length increases the predicted heat loss. Their magnitudes can be compared directly because both input lengths use metres: the same small increase in b has the larger effect.

The local change for b increasing by 0.02 m is

$$\Delta Q \approx Q_b(2,1)\Delta b = 12(0.02) = \boxed{0.24 \text{ W}}.$$

In this particular model the estimate is exact when only b changes: at fixed $a = 2$, $Q(2,b) = 12b + 4$ is linear in b.

Higher partial derivatives and the mixed-derivative theorem

Order of differentiation and the Hessian

The subscript convention is

$$f_{xy} = (f_x)_y, \qquad f_{yx} = (f_y)_x.$$

Thus f_{xy} means first differentiate in x, then in y. The stacked notation for the same operation is $\partial^2 f/(\partial y\,\partial x)$. The four second partials form the Hessian

$$H_f = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}.$$

Mixed Derivative Theorem. If $f, f_x, f_y, f_{xy}, f_{yx}$ are defined throughout an open region containing (a,b) and are all continuous at (a,b), then $f_{xy}(a,b) = f_{yx}(a,b)$. In particular, continuous second partial derivatives on a neighbourhood are sufficient; equality is not an unconditional rule for arbitrary functions.

Example 10. Build the second derivatives from the first derivatives

Let

$$f(x,y) = e^{x-y} + x^2y^3.$$

Find the first partials, all four second partials and the Hessian. Also find f_{xyy} and explain why the mixed second partials agree.

Solution. The first derivatives are

$$f_x = e^{x-y} + 2xy^3, \qquad f_y = -e^{x-y} + 3x^2y^2.$$

The minus sign in f_y comes from differentiating the exponent $x - y$ with respect to y.

Now differentiate each first derivative once more:

$$f_{xx} = \frac{\partial}{\partial x}(e^{x-y} + 2xy^3) = e^{x-y} + 2y^3,$$
$$f_{xy} = \frac{\partial}{\partial y}(e^{x-y} + 2xy^3) = -e^{x-y} + 6xy^2,$$
$$f_{yx} = \frac{\partial}{\partial x}(-e^{x-y} + 3x^2y^2) = -e^{x-y} + 6xy^2,$$
$$f_{yy} = \frac{\partial}{\partial y}(-e^{x-y} + 3x^2y^2) = e^{x-y} + 6x^2y.$$

Therefore

$$\boxed{H_f = \begin{pmatrix} e^{x-y} + 2y^3 & -e^{x-y} + 6xy^2 \\ -e^{x-y} + 6xy^2 & e^{x-y} + 6x^2y \end{pmatrix}.}$$

For the third derivative, start from f_{xy} and differentiate again in y:

$$\boxed{f_{xyy} = \frac{\partial}{\partial y}(-e^{x-y} + 6xy^2) = e^{x-y} + 12xy.}$$

Polynomials and exponential functions have continuous derivatives of all orders here. The theorem's hypotheses hold, so the equality of the two mixed second derivatives is justified, as well as confirmed by calculation.

Example 11. Mixed derivatives can exist and still be unequal

Define

$$f(x,y) = \begin{cases} \dfrac{xy(x^4 - y^2)}{x^4 + y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Show that f is continuous at the origin. Find $f_{xy}(0,0)$ and $f_{yx}(0,0)$ from the definitions, and explain why their values do not contradict the Mixed Derivative Theorem.

Solution.

Continuity is a separate question

Since $|x^4 - y^2| \leq x^4 + y^2$,

$$|f(x,y)| \leq |xy| \leq \frac{x^2 + y^2}{2} \longrightarrow 0.$$

This equals $f(0,0)$, proving continuity.

Find the first partial along each coordinate axis

Fix $y \neq 0$. Because $f(0,y) = 0$,

$$f_x(0,y) = \lim_{h \to 0} \frac{f(h,y) - f(0,y)}{h}$$
$$= \lim_{h \to 0} y \, \frac{h^4 - y^2}{h^4 + y^2} = -y.$$

At $y = 0$, the difference quotient is zero, so $f_x(0,0) = 0$. Thus $f_x(0,y) = -y$ for every y.

Similarly, for $x \neq 0$,

$$f_y(x,0) = \lim_{k \to 0} \frac{f(x,k) - f(x,0)}{k}$$
$$= \lim_{k \to 0} x \, \frac{x^4 - k^2}{x^4 + k^2} = x.$$

At $x = 0$ it is again zero. Hence $f_y(x,0) = x$ for every x.

Differentiate those first-partial values at the origin

Using the stated subscript convention,

$$f_{xy}(0,0) = \lim_{k\to 0} \frac{f_x(0,k) - f_x(0,0)}{k} = \lim_{k\to 0} \frac{-k}{k} = -1,$$

whereas

$$f_{yx}(0,0) = \lim_{h\to 0} \frac{f_y(h,0) - f_y(0,0)}{h} = \lim_{h\to 0} \frac{h}{h} = 1.$$

Thus

$$\boxed{f_{xy}(0,0) = -1 \neq 1 = f_{yx}(0,0).}$$

Identify the failed continuity condition

For $x \neq 0$, the product and quotient rules give

$$f_x(x,y) = y\frac{x^4 - y^2}{x^4 + y^2} + \frac{8x^4y^3}{(x^4 + y^2)^2}.$$

Since $f_x(x,0) = 0$, its y-derivative at $y = 0$ is

$$f_{xy}(x,0) = \lim_{k\to 0}\left[\frac{x^4 - k^2}{x^4 + k^2} + \frac{8x^4k^2}{(x^4 + k^2)^2}\right] = 1.$$

These values remain 1 as $x \to 0$, but $f_{xy}(0,0) = -1$. Therefore f_{xy} is not continuous at the origin. The theorem's hypotheses fail, so there is no contradiction.

Repeated derivatives of a product

Leibniz's rule

If u and v have derivatives through order n, then

$$(uv)^{(n)} = \sum_{r=0}^{n} \binom{n}{r} u^{(r)}v^{(n-r)}, \qquad \binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

A polynomial eventually has zero derivatives, so most terms may disappear. For a trigonometric factor,

$$\frac{\mathrm{d}^k}{\mathrm{d}x^k}\cos x = \cos\left(x + \frac{k\pi}{2}\right).$$

Example 12. A formula for every higher derivative

For $n \geq 2$, find the nth derivative of

$$Y(x) = (x^2 + 1)\cos x.$$

Use the formula to find $Y^{(4)}(0)$.

Solution. Set $u = x^2 + 1$ and $v = \cos x$. Then

$$u' = 2x, \qquad u'' = 2, \qquad u^{(r)} = 0 \quad (r \geq 3).$$

Only the first three Leibniz terms survive:

$$Y^{(n)} = uv^{(n)} + nu'v^{(n-1)} + \binom{n}{2}u''v^{(n-2)}.$$

Substitution gives the complete formula

$$\begin{aligned} Y^{(n)} &= (x^2+1)\cos\left(x+\frac{n\pi}{2}\right) \\ &\quad + 2nx\cos\left(x+\frac{(n-1)\pi}{2}\right) \\ &\quad + n(n-1)\cos\left(x+\frac{(n-2)\pi}{2}\right). \end{aligned}$$

Writing $\alpha = x + n\pi/2$, use $\cos(\alpha - \pi/2) = \sin\alpha$ and $\cos(\alpha - \pi) = -\cos\alpha$. Thus an equivalent compact form is

$$\boxed{Y^{(n)} = [x^2 + 1 - n(n-1)]\cos\left(x+\frac{n\pi}{2}\right) + 2nx\sin\left(x+\frac{n\pi}{2}\right).}$$

At $n = 4, x = 0$, $\cos 2\pi = 1$ and the term containing x vanishes:

$$\boxed{Y^{(4)}(0) = 1 - 4(3) = -11.}$$

Check. For $n = 2$, the formula gives $(1 - x^2)\cos x - 4x\sin x$, which also follows by differentiating $(x^2 + 1)\cos x$ twice directly.

Implicit partial derivatives and differential-equation checks

When the dependent variable is not isolated

If a smooth relation $F(x, y, z) = 0$ defines $z = z(x, y)$, differentiation gives

$$F_x + F_z z_x = 0, \qquad F_y + F_z z_y = 0.$$

At a point where $F_z \neq 0$,

$$\boxed{z_x = -\frac{F_x}{F_z}, \qquad z_y = -\frac{F_y}{F_z}.}$$

The symbols F_x, F_y, F_z treat its three arguments as independent. In the original relation, however, each occurrence of $z(x, y)$ needs a chain-rule factor. Always check that the stated point satisfies the relation and any logarithm or denominator restrictions.

Example 13. A logarithm in an implicit relation

The equation

$$xz + y\ln z = 2, \qquad z > 0,$$

defines z locally as a function of x, y near $(1, 0, 2)$. Find formulas for z_x, z_y and evaluate them at that point.

Solution. First, $1(2) + 0\ln 2 = 2$ and $2 > 0$, so the point is allowed.

Differentiate with respect to x, holding y fixed. The product xz has two varying factors:

$$z + xz_x + y\frac{z_x}{z} = 0.$$

Collect the terms containing z_x:

$$\left(x + \frac{y}{z}\right) z_x = -z.$$

Similarly, differentiation with respect to y gives

$$xz_y + \ln z + y\frac{z_y}{z} = 0, \qquad \left(x + \frac{y}{z}\right) z_y = -\ln z.$$

Therefore, wherever the indicated denominator is nonzero,

$$\boxed{z_x = -\frac{z}{x + y/z}, \qquad z_y = -\frac{\ln z}{x + y/z}.}$$

Here $F_z = x + y/z$, and at $(1, 0, 2)$ it equals $1 \neq 0$. The relation is smooth near the point, so the local formulas apply:

$$\boxed{z_x(1, 0) = -2, \qquad z_y(1, 0) = -\ln 2.}$$

The arguments $(1, 0)$ refer to the independent variables; the corresponding dependent value is $z = 2$.

Two equations checked using second partial derivatives

The two-dimensional Laplace equation is $u_{xx} + u_{yy} = 0$. The one-dimensional wave equation is $v_{tt} = c^2 v_{xx}$, where $c > 0$ is the wave speed. To verify either equation for a supplied smooth function, calculate the required second derivatives and substitute them.

Example 14. Verify a polynomial field and a travelling profile

(a) Show that $u(x, y) = x^3 - 3xy^2 + 2x$ satisfies Laplace's equation.

(b) Find the positive c for which

$$v(x, t) = (x - 2t)^3 + \sin(x + 2t)$$

satisfies the wave equation.

Solution. For (a),

$$u_x = 3x^2 - 3y^2 + 2, \qquad u_{xx} = 6x,$$

and

$$u_y = -6xy, \qquad u_{yy} = -6x.$$

Adding the required derivatives gives

$$\boxed{u_{xx} + u_{yy} = 6x - 6x = 0.}$$

For (b), the spatial derivatives are

$$v_x = 3(x - 2t)^2 + \cos(x + 2t), \qquad v_{xx} = 6(x - 2t) - \sin(x + 2t).$$

In the time derivatives, the inner derivatives are -2 and 2:

$$v_t = -6(x - 2t)^2 + 2\cos(x + 2t),$$

$$v_{tt} = 24(x - 2t) - 4\sin(x + 2t) = 4v_{xx}.$$

Thus $c^2 = 4$. Since a speed is taken to be positive,

$$\boxed{c = 2.}$$

The identity holds for all x, t, not merely at a selected point.

Differentiability and local approximation

What differentiability requires

The existence of $f_x(a,b)$ and $f_y(a,b)$ gives two slopes. Differentiability requires these slopes to describe the change in f for *every* small displacement. More precisely, write

$$f(a+h,b+k)-f(a,b)=f_x(a,b)h+f_y(a,b)k+R(h,k).$$

The function is differentiable at (a,b) if

$$\lim_{(h,k)\to(0,0)}\frac{R(h,k)}{\sqrt{h^2+k^2}}=0.$$

Thus the error in the linear approximation must become small compared with the length of the displacement.
If the first partial derivatives exist on an open neighbourhood and are continuous there, the function is differentiable. Differentiability implies continuity. The existence of partial derivatives alone implies neither.

Example 15. Partial derivatives do not settle differentiability

Each function below is assigned the value 0 at $(0,0)$:

$$H(x,y)=\frac{x^2y}{(x^2+y^2)^{3/2}},\qquad G(x,y)=\frac{xy}{|x|+|y|}\qquad((x,y)\neq(0,0)).$$

For each function, find the first partial derivatives at the origin and decide whether it is continuous and differentiable there.

Solution.

The partial derivatives only examine the coordinate directions.

Both functions are zero on the two axes. For example,

$$H_x(0,0)=\lim_{h\to0}\frac{H(h,0)-H(0,0)}{h}=\lim_{h\to0}0=0.$$

The same calculation on the other axis, and for G, gives

$$H_x(0,0)=H_y(0,0)=G_x(0,0)=G_y(0,0)=0.$$

The first function is not continuous.

On the path $x=y=t$ with $t>0$,

$$H(t,t)=\frac{t^3}{(2t^2)^{3/2}}=\frac{1}{2\sqrt{2}}.$$

This does not tend to $H(0,0)=0$. Hence H is not continuous at the origin. Because differentiability would imply continuity, it is also **not differentiable at the origin**.

The second function is continuous.

For $(x,y)\neq(0,0)$,

$$|G(x,y)|=\frac{|x|\,|y|}{|x|+|y|}\le|x|\le\sqrt{x^2+y^2}.$$

The bound tends to zero, so $G(x,y)\to0=G(0,0)$.

Continuity still does not give differentiability.

Since both first partials of G at the origin are zero, its proposed linear part is zero. Its remainder is therefore $R(h,k)=G(h,k)$. Along $h=k=t>0$,

$$G(t,t)=\frac{t^2}{2t}=\frac{t}{2},\qquad\frac{R(t,t)}{\sqrt{t^2+t^2}}=\frac{t/2}{\sqrt{2}\,t}=\frac{1}{2\sqrt{2}}.$$

The required remainder ratio does not approach zero. Thus

$$\boxed{G \text{ is continuous, but not differentiable, at } (0,0).}$$

Linearisation of a function of two variables

For a differentiable function f near (a,b), its linearisation is

$$L(x,y) = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b).$$

The approximation $f(x,y) \approx L(x,y)$ is useful for (x,y) close to (a,b). The plane $z = L(x,y)$ is the tangent plane to the graph $z = f(x,y)$ at $(a,b,f(a,b))$. The formula for L is exact; using it in place of f is generally an approximation.

Example 16. Approximate a square root and identify its tangent plane

For

$$f(x,y) = \sqrt{x+y^2},$$

find the linearisation at $(5,2)$, use it to estimate $f(5.06, 1.97)$, and give the tangent plane at the corresponding point of the graph.

Solution. At the base point, $x + y^2 = 5 + 4 = 9 > 0$. The first partials are continuous on the open region $x + y^2 > 0$, so linearisation is justified near $(5,2)$. Differentiating gives

$$f_x = \frac{1}{2\sqrt{x+y^2}}, \qquad f_y = \frac{2y}{2\sqrt{x+y^2}} = \frac{y}{\sqrt{x+y^2}}.$$

Thus

$$f(5,2) = 3, \qquad f_x(5,2) = \frac{1}{6}, \qquad f_y(5,2) = \frac{2}{3}.$$

Substituting these three values,

$$\boxed{L(x,y) = 3 + \frac{x-5}{6} + \frac{2}{3}(y-2).}$$

The changes in the inputs are $5.06 - 5 = 0.06$ and $1.97 - 2 = -0.03$. Hence

$$\begin{aligned} f(5.06, 1.97) &\approx L(5.06, 1.97) \\ &= 3 + \frac{0.06}{6} + \frac{2}{3}(-0.03) \\ &= 3 + 0.01 - 0.02 = \boxed{2.99}. \end{aligned}$$

The graph point is $(5,2,3)$, so the tangent plane is

$$\boxed{z = 3 + \frac{x-5}{6} + \frac{2}{3}(y-2).}$$

Check. At the base point, the plane gives $z = 3$. The input x increases slightly, which raises the estimate by 0.01; y decreases slightly, which lowers it by 0.02. Their net effect is a decrease of 0.01.

Linearisation with three inputs

If f is differentiable at (a,b,c), then

$$\begin{aligned} L(x,y,z) = f(a,b,c) &+ f_x(a,b,c)(x-a) \\ &+ f_y(a,b,c)(y-b) + f_z(a,b,c)(z-c). \end{aligned}$$

Each coefficient measures the effect of changing one input while keeping the other two fixed. Continuous first partials near the base point are a sufficient condition for using this formula.

Example 17. Three small input changes in a quotient

Find the linearisation of

$$f(x,y,z)=\frac{x}{y+z}$$

at $(6,1,2)$, and estimate $f(6.03,0.99,2.02)$.

Solution. The denominator is $1+2=3\neq 0$ at the base point. It remains nonzero on a sufficiently small neighbourhood, where all first partials are continuous. We have

$$f_x=\frac{1}{y+z},\qquad f_y=-\frac{x}{(y+z)^2},\qquad f_z=-\frac{x}{(y+z)^2}.$$

At $(6,1,2)$ these give

$$f=2,\qquad f_x=\frac{1}{3},\qquad f_y=f_z=-\frac{2}{3}.$$

Therefore

$$\boxed{L(x,y,z)=2+\frac{x-6}{3}-\frac{2}{3}(y-1)-\frac{2}{3}(z-2).}$$

The input changes are $0.03,-0.01,0.02$, respectively. Thus

$$\begin{aligned}f(6.03,0.99,2.02)&\approx 2+\frac{0.03}{3}-\frac{2}{3}(-0.01)-\frac{2}{3}(0.02)\\&=2+\frac{1}{100}+\frac{1}{150}-\frac{1}{75}\\&=\boxed{\frac{601}{300}\approx 2.00333}.\end{aligned}$$

The two denominator changes partly offset each other; both must still be included in the linearisation.

Differentials and measurement errors

An exact change and its first-order estimate

At a fixed base point (a,b), the total differential is

$$\mathrm{d}f=f_x(a,b)\,\mathrm{d}x+f_y(a,b)\,\mathrm{d}y.$$

To estimate the effect of input changes $\Delta x,\Delta y$, set $\mathrm{d}x=\Delta x,\ \mathrm{d}y=\Delta y$. Then

$$\Delta f=f(a+\Delta x,b+\Delta y)-f(a,b)\approx \mathrm{d}f.$$

The signs of $\mathrm{d}x$, $\mathrm{d}y$ matter when the changes are known. When only error magnitudes are known, the triangle inequality gives the first-order estimate

$$|\Delta f|\lesssim |f_x(a,b)|\,|\Delta x|+|f_y(a,b)|\,|\Delta y|.$$

Here $\lesssim$ indicates an approximate bound obtained from the linearisation, rather than an exact bound on the nonlinear error.

Example 18. Compare a differential with the actual change

Let $f(x, y) = \ln(2x - y)$. As (x, y) changes from $(2, 1)$ to $(2.02, 1.01)$, find the differential estimate of the change in f and compare it with the exact change.

Solution. Both points lie in the domain $2x - y > 0$: their logarithm arguments are 3 and 3.03. The first partials are

$$f_x = \frac{2}{2x - y}, \qquad f_y = -\frac{1}{2x - y}.$$

At the initial point, $f_x = 2/3$ and $f_y = -1/3$. Taking the signed changes $\mathrm{d}x = 0.02$ and $\mathrm{d}y = 0.01$, we obtain

$$\mathrm{d}f = \frac{2}{3}(0.02) - \frac{1}{3}(0.01) = \frac{0.04 - 0.01}{3} = \boxed{0.01}.$$

The estimate predicts an increase of approximately 0.01.

The exact change follows directly from the two function values:

$$\begin{aligned} \Delta f &= \ln(2(2.02) - 1.01) - \ln(2(2) - 1) \\ &= \ln(3.03) - \ln 3 = \ln\left(\frac{3.03}{3}\right) \\ &= \boxed{\ln(1.01) \approx 0.00995033}. \end{aligned}$$

The differential is close to the exact change, but is not equal to it.

Example 19. Changes and uncertainty in a wire's resistance

For a uniform wire, use the model

$$R = \frac{\rho L}{\pi a^2},$$

where $L > 0$ is its length, $a > 0$ its radius and the resistivity $\rho > 0$ is fixed. Lengths are measured in metres and R in ohms.

(a) Derive a formula for the approximate relative change in R.

(b) Estimate the percentage change in R if L increases by 0.4% and a increases by 0.3%.

(c) If the measurement errors have unknown signs but magnitudes at most 0.4% in L and 0.3% in a, estimate the largest possible absolute error to first order when the nominal resistance is $10\,\Omega$.

Solution.

Differentiate before substituting percentage changes.

Since ρ is constant,

$$R_L = \frac{\rho}{\pi a^2} = \frac{R}{L}, \qquad R_a = -\frac{2\rho L}{\pi a^3} = -\frac{2R}{a}.$$

Therefore

$$\mathrm{d}R = \frac{R}{L}\,\mathrm{d}L - \frac{2R}{a}\,\mathrm{d}a, \qquad \boxed{\frac{\mathrm{d}R}{R} = \frac{\mathrm{d}L}{L} - 2\frac{\mathrm{d}a}{a}.}$$

The ratios are dimensionless. For small changes, $\Delta R/R$ is approximately this relative differential.

Known increases have known signs.

The given increases mean $\mathrm{d}L/L = 0.004$ and $\mathrm{d}a/a = 0.003$. Hence

$$\frac{\mathrm{d}R}{R} = 0.004 - 2(0.003) = -0.002.$$

Multiplying by 100 converts this to a percentage:

$$\boxed{\text{The resistance decreases by approximately } 0.2\%.}$$

The increase in radius has the larger effect because it enters the denominator as a square.

Unknown error signs require a magnitude estimate.

For the uncertainty calculation, cancellation cannot be assumed:

$$\left|\frac{\mathrm{d}R}{R}\right| \le \left|\frac{\mathrm{d}L}{L}\right| + 2\left|\frac{\mathrm{d}a}{a}\right| \le 0.004 + 2(0.003) = 0.010.$$

Thus the largest relative error is estimated to first order as 1.0%. For a nominal resistance of $10\,\Omega$, the corresponding absolute error estimate is

$$\boxed{10(0.010) = 0.10\,\Omega.}$$

The 1.0% uncertainty estimate and the -0.2% signed change answer different questions.

The chain rule

Several inputs changing with one parameter

If $W = W(x, y, z)$ is differentiable, and $x(t), y(t), z(t)$ are differentiable, then their composition satisfies

$$\frac{\mathrm{d}W}{\mathrm{d}t} = W_x\frac{\mathrm{d}x}{\mathrm{d}t} + W_y\frac{\mathrm{d}y}{\mathrm{d}t} + W_z\frac{\mathrm{d}z}{\mathrm{d}t}.$$

All partial derivatives are evaluated at the moving point $(x(t), y(t), z(t))$. Each term accounts for one way that t changes W. For a function of two inputs, omit the third term.

Example 20. Three contributions to a rate of change

Let

$$W(x, y, z) = xy + \ln z, \qquad x = t^2, \quad y = e^t, \quad z = 1 + t \quad (t > -1).$$

Find $\mathrm{d}W/\,\mathrm{d}t$, evaluate it at $t = 1$, and check by direct substitution.

Solution. The restriction $t > -1$ ensures $z > 0$, so the logarithm and its derivatives are defined. First differentiate W with its other inputs held fixed:

$$W_x = y, \qquad W_y = x, \qquad W_z = \frac{1}{z}.$$

The input rates are

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 2t, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = e^t, \qquad \frac{\mathrm{d}z}{\mathrm{d}t} = 1.$$

Substitution in the chain rule gives

$$\begin{aligned}\frac{\mathrm{d}W}{\mathrm{d}t} &= y(2t) + x(e^t) + \frac{1}{z}(1)\\ &= \boxed{2te^t + t^2e^t + \frac{1}{1+t}}.\end{aligned}$$

At $t = 1$,

$$\boxed{\left.\frac{\mathrm{d}W}{\mathrm{d}t}\right|_{t=1} = 3e + \frac{1}{2}.}$$

For an independent check, express the whole composition in terms of t:

$$W(t) = t^2e^t + \ln(1 + t).$$

The ordinary product and chain rules give

$$W'(t) = 2te^t + t^2e^t + \frac{1}{1+t},$$

which agrees with the sum of the three contributions.

Two independent parameters

For a differentiable function $z = f(x, y)$ with differentiable inputs $x = x(u, v)$ and $y = y(u, v)$,

$$z_u = f_x\, x_u + f_y\, y_u, \qquad z_v = f_x\, x_v + f_y\, y_v.$$

In z_u, the parameter v is held fixed. In z_v, the parameter u is held fixed. The partials f_x, f_y are evaluated after substituting the input functions.

Example 21. The same output with two independent parameters

Let

$$z = \ln(x - y), \qquad x = u^2 + v, \qquad y = uv.$$

Find z_u and z_v, and evaluate both at $(u, v) = (2, 1)$.

Solution. The composition is defined where $u^2 + v - uv > 0$. At $(2, 1)$, the input point is $(x, y) = (5, 2)$ and $x - y = 3 > 0$, so the derivative formulas apply.

The outer and inner derivatives are

$$z_x = \frac{1}{x - y}, \qquad z_y = -\frac{1}{x - y},$$

$$x_u = 2u, \quad y_u = v, \qquad x_v = 1, \quad y_v = u.$$

Holding v fixed for the first rate,

$$z_u = z_x\, x_u + z_y\, y_u = \frac{2u - v}{x - y} = \boxed{\frac{2u - v}{u^2 + v - uv}}.$$

Holding u fixed for the second rate,

$$z_v = z_x\, x_v + z_y\, y_v = \frac{1 - u}{x - y} = \boxed{\frac{1 - u}{u^2 + v - uv}}.$$

Consequently,

$$\boxed{z_u(2, 1) = \frac{4 - 1}{3} = 1, \qquad z_v(2, 1) = \frac{1 - 2}{3} = -\frac{1}{3}.}$$

Directly differentiating $z = \ln(u^2 + v - uv)$ gives the same two expressions. Increasing u near the stated point raises z, whereas increasing v with u fixed lowers z.

Local maxima, minima and saddle points

Critical points and the second derivative test

A local minimum at (a, b) means $f(x, y) \geq f(a, b)$ for all sufficiently nearby points of the domain. A local maximum reverses this inequality. A stationary point (one where $f_x = f_y = 0$) is a saddle point if values both greater and smaller than $f(a, b)$ occur arbitrarily close to (a, b).

An interior point is a critical point if both first partial derivatives are zero, or if at least one first partial derivative does not exist. For a differentiable function, an interior local extremum must therefore satisfy $f_x = f_y = 0$. This condition identifies candidates, not their type.

Suppose the first and second partial derivatives are continuous on a neighbourhood of a point where $f_x = f_y = 0$. Calculate at that point

$$D = f_{xx} f_{yy} - (f_{xy})^2.$$

Derivative information	Conclusion
$D > 0$ and $f_{xx} > 0$	Strict local minimum
$D > 0$ and $f_{xx} < 0$	Strict local maximum
$D < 0$	Saddle point
$D = 0$	Test gives no conclusion

When the test gives no conclusion, use the function itself or examine appropriate nearby paths.

Example 22. One function with three different types of critical point

Find and classify all critical points of

$$f(x, y) = x^3 - 3x + y^3 - 3y.$$

Decide whether any local extremum is also an absolute extremum on $\mathbb{R}^2$.

Solution. The function is a polynomial, so its first and second partials are continuous everywhere. Its first partial derivatives are

$$f_x = 3x^2 - 3, \qquad f_y = 3y^2 - 3.$$

The equations $f_x = f_y = 0$ give $x = \pm 1$ and $y = \pm 1$. All four combinations are critical points:

$$(1, 1), \quad (-1, -1), \quad (1, -1), \quad (-1, 1).$$

For classification,

$$f_{xx} = 6x, \qquad f_{yy} = 6y, \qquad f_{xy} = 0, \qquad D = 36xy.$$

Substituting each point gives

Point	D	f_{xx}	Type	Value of f
$(1, 1)$	36	6	Local minimum	-4
$(-1, -1)$	36	-6	Local maximum	4
$(1, -1)$	-36	6	Saddle	0
$(-1, 1)$	-36	-6	Saddle	0

The extrema in the table are local. To check whether absolute extrema exist, keep $y = 0$:

$$f(x, 0) = x^3 - 3x \longrightarrow \begin{cases} +\infty, & x \to +\infty, \\ -\infty, & x \to -\infty. \end{cases}$$

The function is unbounded above and below on $\mathbb{R}^2$. Therefore it has **no absolute maximum or absolute minimum on** $\mathbb{R}^2$.

Example 23. When the second derivative test cannot decide

Classify the origin for each function, giving a reason that does not depend on a conclusive second derivative test:

(a) $p(x, y) = x^4 + y^6$,
(b) $q(x, y) = x^4 - y^6$,
(c) $h(x, y) = |x| + y^2$.

Solution. For (a) and (b), the first partials are $4x^3$ and $\pm 6y^5$, so both vanish at the origin. The second partials are $12x^2$, $\pm 30y^4$ and the mixed partial 0. All are zero at the origin. Thus $D = 0$ for both functions, and the test cannot distinguish them.

(a) Both terms are nonnegative.

For every (x, y),

$$p(x, y) = x^4 + y^6 \geq 0 = p(0, 0).$$

Equality requires both $x = 0$ and $y = 0$. Hence the origin is a **strict absolute minimum, with value** 0.

(b) Two simple paths give opposite signs.

On the x-axis,

$$q(x, 0) = x^4 > 0 \qquad (x \neq 0).$$

On the y-axis,

$$q(0, y) = -y^6 < 0 \qquad (y \neq 0).$$

Both paths contain points arbitrarily close to the origin. Therefore the origin is a **saddle point**.

(c) A minimum can occur where a partial derivative is undefined.

The two summands in $h(x, y)$ are nonnegative, and their sum is zero only at the origin. Thus the origin is again a strict absolute minimum. However,

$$\frac{h(t, 0) - h(0, 0)}{t} = \frac{|t|}{t} = \begin{cases} 1, & t > 0, \\ -1, & t < 0. \end{cases}$$

The two one-sided limits disagree, so $h_x(0, 0)$ does not exist. The smooth second derivative test does not apply. The direct nonnegativity argument proves the minimum without that test.

Absolute extrema on a region

Include the boundary

A continuous function on a nonempty closed, bounded region attains an absolute maximum and an absolute minimum. To find them:

1. Find all interior critical points, including any where a first partial derivative is undefined.
2. Restrict the function to each boundary curve. Find its extrema over the corresponding interval, including endpoints and corners.
3. Compare the function values at all candidates.

The equations $f_x = f_y = 0$ alone do not examine the boundary. A curved boundary can often be handled by substituting its defining relation, reducing the calculation to one variable.

Example 24. A rectangle requires four boundary checks

Find the absolute maximum and minimum of

$$f(x, y) = x^2 + xy + y^2 - 3x - 4y$$

on the rectangle $0 \leq x \leq 2$, $0 \leq y \leq 3$.

Solution. The function is continuous and the rectangle is closed and bounded, so both absolute extrema exist.

Interior critical point.

The first partials are

$$f_x = 2x + y - 3, \qquad f_y = x + 2y - 4.$$

Solving the simultaneous equations gives

$$2x + y = 3, \quad x + 2y = 4 \quad \Longrightarrow \quad x = \frac{2}{3}, \quad y = \frac{5}{3}.$$

This point lies in the interior. Its value is

$$f\left(\frac{2}{3}, \frac{5}{3}\right) = \frac{4}{9} + \frac{10}{9} + \frac{25}{9} - 2 - \frac{20}{3} = -\frac{13}{3}.$$

Left boundary: $x = 0$, with $0 \le y \le 3$.

The restricted function and derivative are

$$f(0, y) = y^2 - 4y, \qquad \frac{\mathrm{d}}{\mathrm{d}y} f(0, y) = 2y - 4.$$

The stationary value occurs at $y = 2$, which lies in this interval. The candidates on this edge are $y = 0, 2, 3$.

Right boundary: $x = 2$, with $0 \le y \le 3$.

Here

$$f(2, y) = y^2 - 2y - 2, \qquad \frac{\mathrm{d}}{\mathrm{d}y} f(2, y) = 2y - 2.$$

The stationary point is $y = 1$. The candidates are $y = 0, 1, 3$.

Lower boundary: $y = 0$, with $0 \le x \le 2$.

Here

$$f(x, 0) = x^2 - 3x, \qquad \frac{\mathrm{d}}{\mathrm{d}x} f(x, 0) = 2x - 3.$$

The stationary point $x = 3/2$ belongs to the interval. The candidates are $x = 0, 3/2, 2$.

Upper boundary: $y = 3$, with $0 \le x \le 2$.

Substitution gives

$$f(x, 3) = x^2 - 3, \qquad \frac{\mathrm{d}}{\mathrm{d}x} f(x, 3) = 2x.$$

The derivative is zero only at $x = 0$, an endpoint already included. There is no further interior candidate on this edge; its endpoints are $x = 0$ and $x = 2$.

Compare the complete candidate list.

Repeated corners need be listed only once:

Candidate	Reason for inclusion	Value of f
$(2/3, 5/3)$	Interior stationary point	$-13/3$
$(0, 2)$	Left-edge stationary point	-4
$(2, 1)$	Right-edge stationary point	-3
$(3/2, 0)$	Lower-edge stationary point	$-9/4$
$(0, 0)$	Corner	0
$(2, 0)$	Corner	-2
$(0, 3)$	Corner	-3
$(2, 3)$	Corner	1

The smallest and largest entries give

$$\boxed{\min f = -\frac{13}{3} \text{ at } \left(\frac{2}{3}, \frac{5}{3}\right), \qquad \max f = 1 \text{ at } (2, 3).}$$

Example 25. Absolute extrema when no interior stationary point exists

Find the absolute extrema of

$$f(x, y) = x + y^2$$

on the closed disk $x^2 + y^2 \le 1$.

Solution. The polynomial is continuous and the disk is closed and bounded, so absolute extrema exist. In the interior,

$$f_x = 1, \qquad f_y = 2y.$$

The equation $f_x = 0$ has no solution. Both partials exist everywhere, so there are no interior critical points. The extrema must therefore occur on the circle $x^2 + y^2 = 1$.

On this circle, $y^2 = 1 - x^2$ and $-1 \le x \le 1$. Each value of x in this interval corresponds to at least one point on the circle. Thus the boundary values are exactly the values of

$$F(x) = x + (1 - x^2) = 1 + x - x^2, \qquad -1 \le x \le 1.$$

Since

$$F'(x) = 1 - 2x = 0 \quad \Longrightarrow \quad x = \frac{1}{2},$$

the candidates are $x = -1, 1/2, 1$. Their values are

$$F(-1) = -1, \qquad F\left(\frac{1}{2}\right) = \frac{5}{4}, \qquad F(1) = 1.$$

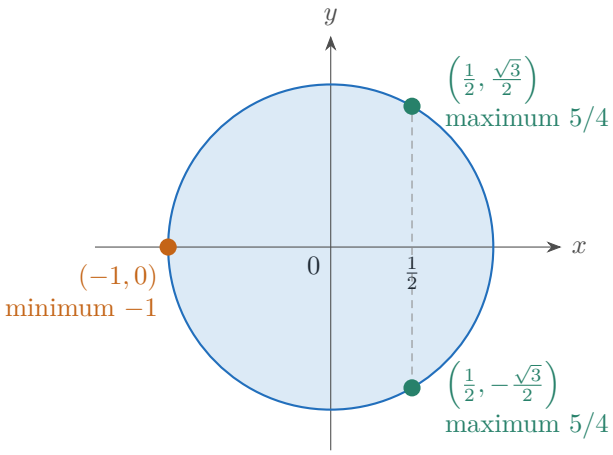
At $x = -1$, the circle equation gives $y = 0$. At $x = 1/2$, it gives

$$y^2 = 1 - \frac{1}{4} = \frac{3}{4} \quad \Longrightarrow \quad y = \pm\frac{\sqrt{3}}{2}.$$

Consequently,

$$\boxed{\min f = -1 \text{ at } (-1, 0),}$$

$$\boxed{\max f = \frac{5}{4} \text{ at } \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \text{ and } \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right).}$$



Both boundary points with $x = 1/2$ must be retained: the function depends on y^2, so they give the same maximum.

An optimisation model

Choose variables, express the restriction, then minimise

An optimisation model needs an objective function and a domain. A restriction such as a fixed volume can eliminate one variable before differentiation. A second derivative test establishes a local result; a separate argument may be needed to prove that the result is the best over the whole domain.
One useful inequality for three positive numbers is

$$A + B + C \geq 3\sqrt[3]{ABC}.$$

Equality holds precisely when $A = B = C$. This is the arithmetic–geometric mean inequality.

Example 26. Use the least material for an open container

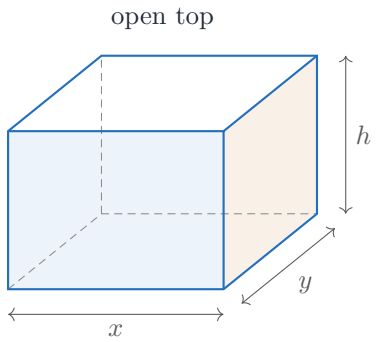
A rectangular container has no lid and must hold $4\,\mathrm{m}^3$. The base and four sides use the same material at the same cost per square metre. Ignore material thickness and waste. Find the dimensions that use the least material, and justify that the minimum is global.

Solution. Let $x, y > 0$ be the base dimensions and $h > 0$ the height, all in metres. The volume condition gives

$$xyh = 4 \quad \Longrightarrow \quad h = \frac{4}{xy}.$$

There is one base of area xy, two sides of area xh and two sides of area yh. There is no top face to include. Hence the material area is

$$S = xy + 2xh + 2yh = xy + 2x\left(\frac{4}{xy}\right) + 2y\left(\frac{4}{xy}\right) = \boxed{xy + \frac{8}{x} + \frac{8}{y}}, \qquad x, y > 0.$$



Find the stationary dimensions.

Differentiate on the positive domain:

$$S_x = y - \frac{8}{x^2}, \qquad S_y = x - \frac{8}{y^2}.$$

Setting both to zero gives

$$x^2 y = 8, \qquad xy^2 = 8.$$

Since $x, y > 0$, division of these equations gives $x/y = 1$, so $x = y$. Then $x^3 = 8$, hence

$$x = y = 2, \qquad h = \frac{4}{2 \cdot 2} = 1.$$

Check the local classification.

The second derivatives are

$$S_{xx} = \frac{16}{x^3}, \qquad S_{yy} = \frac{16}{y^3}, \qquad S_{xy} = 1.$$

At $(2,2)$, $S_{xx} = S_{yy} = 2$ and

$$D = 2 \cdot 2 - 1^2 = 3 > 0.$$

Since $S_{xx} > 0$, these dimensions give a strict local minimum.

Show that no other positive dimensions use less material.

The three positive terms $xy, 8/x, 8/y$ have the constant product

$$(xy)\left(\frac{8}{x}\right)\left(\frac{8}{y}\right) = 64.$$

The arithmetic–geometric mean inequality therefore gives

$$S = xy + \frac{8}{x} + \frac{8}{y} \geq 3\sqrt[3]{64} = 12.$$

At $x = y = 2$, all three terms equal 4, so equality is attained. Thus this is a global minimum over every permissible container:

$$\boxed{\text{base } 2\,\text{m} \times 2\,\text{m}, \quad \text{height } 1\,\text{m}, \quad \text{minimum area } 12\,\text{m}^2.}$$

Check. The volume is $2 \cdot 2 \cdot 1 = 4\,\text{m}^3$. The base uses $4\,\text{m}^2$ and the four sides together use $8\,\text{m}^2$, giving the same total $12\,\text{m}^2$.

Arc length: adding small distances

The three useful forms of ds

A small displacement along a plane curve has length $ds = \sqrt{dx^2 + dy^2}$. Adding these lengths gives the arc length. For continuously differentiable coordinate functions, the formulas are

$$\begin{aligned}
&y = f(x), \quad a \leq x \leq b: && L = \int_a^b \sqrt{1 + [f'(x)]^2}\,\mathrm{d}x, \\
&x = g(y), \quad c \leq y \leq d: && L = \int_c^d \sqrt{1 + [g'(y)]^2}\,\mathrm{d}y, \\
&x = x(t),\ y = y(t), \quad \alpha \leq t \leq \beta: && L = \int_\alpha^\beta \sqrt{[x'(t)]^2 + [y'(t)]^2}\,\mathrm{d}t.
\end{aligned}$$

The parameter interval must trace the required arc once. A curve made of finitely many such pieces is handled by adding their lengths. In every form, the quantity under the square root is a sum of squares; length is nonnegative.

Evaluating a definite integral

If g is continuous on $[a,b]$ and $G' = g$, the Fundamental Theorem of Calculus gives

$$\int_a^b g(t)\,\mathrm{d}t = G(b) - G(a).$$

For a substitution $u = \phi(t)$, change the differential and the bounds together: $t = a, b$ become $u = \phi(a), \phi(b)$. An antiderivative in u must be evaluated using u-bounds.

Example 27. Arc length when y is given in terms of x

Find the length of

$$y = \sqrt{2x - x^2}, \qquad \frac{1}{2} \le x \le \frac{3}{2}.$$

Solution. The expression under the root is positive throughout this interval, so the derivative is defined there. Differentiate before forming the length integrand:

$$y' = \frac{2 - 2x}{2\sqrt{2x - x^2}} = \frac{1 - x}{\sqrt{2x - x^2}}.$$

Combine the terms over one denominator:

$$1 + (y')^2 = \frac{2x - x^2 + (1 - x)^2}{2x - x^2} = \frac{1}{2x - x^2} = \frac{1}{1 - (x - 1)^2}.$$

Using $u = x - 1$, we have $du = dx$, and the new limits are $-\frac{1}{2}$ and $\frac{1}{2}$. The standard formula $\int (1 - u^2)^{-1/2}\, du = \sin^{-1} u + C$ gives

$$\begin{aligned} L &= \int_{1/2}^{3/2} \frac{1}{\sqrt{1 - (x - 1)^2}}\, dx \\ &= \int_{-1/2}^{1/2} \frac{1}{\sqrt{1 - u^2}}\, du = \left[\sin^{-1} u\right]_{-1/2}^{1/2} = \frac{\pi}{6} - \left(-\frac{\pi}{6}\right). \end{aligned}$$

$$\boxed{L = \frac{\pi}{3}}$$

Check. The curve lies on $(x - 1)^2 + y^2 = 1$. The selected upper arc subtends an angle $\pi/3$ at the centre, so radius times angle also gives $\pi/3$.

Example 28. Arc length when x is given in terms of y

Find the length of

$$x = \frac{y^2}{8} - \ln y, \qquad 2 \le y \le 4.$$

Solution. There is no need to solve this equation for y. Since $y > 0$,

$$\frac{dx}{dy} = \frac{y}{4} - \frac{1}{y}.$$

The extra 1 in the arc-length formula completes a square:

$$\begin{aligned} 1 + \left(\frac{y}{4} - \frac{1}{y}\right)^2 &= 1 + \frac{y^2}{16} - \frac{1}{2} + \frac{1}{y^2} \\ &= \frac{y^2}{16} + \frac{1}{2} + \frac{1}{y^2} = \left(\frac{y}{4} + \frac{1}{y}\right)^2. \end{aligned}$$

The expression $y/4 + 1/y$ is positive on $[2, 4]$, so its square root is $y/4 + 1/y$, with no sign change. Therefore

$$\begin{aligned} L &= \int_2^4 \left(\frac{y}{4} + \frac{1}{y}\right) dy = \left[\frac{y^2}{8} + \ln y\right]_2^4 \\ &= (2 + \ln 4) - \left(\frac{1}{2} + \ln 2\right). \end{aligned}$$

$$\boxed{L = \frac{3}{2} + \ln 2}$$

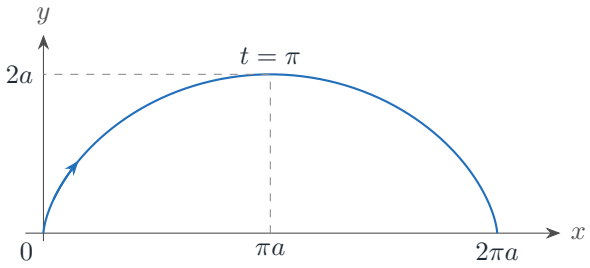
Example 29. The length of one parametrised arch

A point on a rolling circle of radius $a > 0$ follows the curve

$$x = a(t - \sin t), \qquad y = a(1 - \cos t), \qquad 0 \le t \le 2\pi.$$

Find the length of this one arch.

Solution. At $t = 0$ the point is $(0, 0)$; at $t = \pi$ it is $(\pi a, 2a)$; at $t = 2\pi$ it is $(2\pi a, 0)$. Moreover, $x' = a(1 - \cos t) \ge 0$, with strict positivity between the endpoints, so the interval describes the arch once from left to right.



Differentiate both coordinates and simplify using $1 - \cos t = 2\sin^2(t/2)$:

$$x' = a(1 - \cos t), \qquad y' = a \sin t,$$
$$(x')^2 + (y')^2 = a^2\big((1 - \cos t)^2 + \sin^2 t\big) = 2a^2(1 - \cos t) = 4a^2 \sin^2(t/2).$$

Here $0 \le t/2 \le \pi$, so $\sin(t/2) \ge 0$. Thus

$$L = \int_0^{2\pi} 2a \sin(t/2)\, \mathrm{d}t = \big[-4a \cos(t/2)\big]_0^{2\pi}$$
$$= -4a \cos \pi + 4a \cos 0 = 8a.$$

$\boxed{L = 8a}$

The zero speed at the two endpoints does not prevent this length calculation.

Quadric surfaces and their traces

Read a surface by cutting it with planes

A **trace** is the intersection of a surface with a plane such as $x = k$, $y = k$ or $z = k$. Substitute the fixed value into the surface equation and identify the resulting plane curve. A missing coordinate is free to vary: for example, an equation involving only x, z describes a cylinder with straight generating lines parallel to the y-axis.

The table gives six standard families, with $a, b, c > 0$. Permuting the coordinate names changes the direction of the axis. Replacing x, y, z by $x - x_0, y - y_0, z - z_0$ translates the surface.

Surface	Standard equation	Traces that identify it
Ellipsoid	$\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}+\dfrac{z^2}{c^2}=1$	$z=k$: ellipses for $\lvert k\rvert<c$, points at $k=\pm c$, empty for $\lvert k\rvert>c$. All three central coordinate traces are ellipses.
Elliptic cone	$\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=\dfrac{z^2}{c^2}$	$z=k\neq 0$: ellipses. $z=0$: a point. A vertical plane through the axis gives two intersecting lines.
Hyperboloid of one sheet	$\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}-\dfrac{z^2}{c^2}=1$	$z=k$: an ellipse for every k. The central vertical traces are hyperbolas. Axis: the coordinate with the negative square.
Hyperboloid of two sheets	$\dfrac{z^2}{c^2}-\dfrac{x^2}{a^2}-\dfrac{y^2}{b^2}=1$	$z=k$: empty if $\lvert k\rvert<c$, a point if $\lvert k\rvert=c$, an ellipse if $\lvert k\rvert>c$. Axis: the coordinate with the positive square.
Elliptic paraboloid	$z=\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}$	$z=k$: ellipses for $k>0$, a point for $k=0$, empty for $k<0$. Vertical traces are parabolas opening in the same direction.
Hyperbolic paraboloid	$z=\dfrac{x^2}{a^2}-\dfrac{y^2}{b^2}$	$z=k\neq 0$: hyperbolas; $z=0$: intersecting lines. The two central vertical parabolas open in opposite directions.

A sphere is the ellipsoid with $a=b=c$. Cylinders can be circular, elliptic, parabolic or hyperbolic; a cylinder need not be round.

Example 30. Complete squares, then check the traces

(a) Identify the surface

$$4x^2-y^2+4z^2-8x-4y+16z+12=0.$$

Find its centre, axis direction, and traces in the planes $y=-2$, $y=0$ and $x=1$.

(b) Identify $z^2=4x$. Describe its traces when $y=k$ and when $x=k$.

Solution.

(a) Put the equation in standard form

Group each variable with its linear term:

$$\begin{aligned}4(x^2-2x)-(y^2+4y)+4(z^2+4z)+12&=0,\\ 4\big((x-1)^2-1\big)-\big((y+2)^2-4\big)+4\big((z+2)^2-4\big)+12&=0,\\ 4(x-1)^2-(y+2)^2+4(z+2)^2&=4.\end{aligned}$$

Dividing by 4 gives

$$(x-1)^2+(z+2)^2-\frac{(y+2)^2}{4}=1.$$

There are two positive squares and one negative square on the left, with a positive constant on the right. This is a **hyperboloid of one sheet**, centred at $(1,-2,-2)$, with axis parallel to the y-axis.

The requested traces give independent checks:

$$\begin{aligned}y=-2:&\quad (x-1)^2+(z+2)^2=1, &&\text{circle of radius } 1,\\ y=0:&\quad (x-1)^2+(z+2)^2=2, &&\text{circle of radius } \sqrt{2},\\ x=1:&\quad (z+2)^2-\frac{(y+2)^2}{4}=1, &&\text{hyperbola.}\end{aligned}$$

More generally, the trace at every real $y = k$ is

$$(x-1)^2 + (z+2)^2 = 1 + \frac{(k+2)^2}{4}.$$

Its radius is always positive. There is no interval of empty traces separating two sheets.

(b) Recognise the free coordinate

In $z^2 = 4x$, the coordinate y is absent. For *every* fixed $y = k$, the trace is the same parabola $x = z^2/4$, opening in the positive x-direction. The surface is a **parabolic cylinder parallel to the y-axis**. For a plane $x = k$,

$$z^2 = 4k \quad \Longrightarrow \quad \begin{cases} z = \pm 2\sqrt{k},\ y \text{ arbitrary}, & k > 0, \\ z = 0,\ y \text{ arbitrary}, & k = 0, \\ \text{no points}, & k < 0. \end{cases}$$

Thus for $k > 0$ the traces are two straight lines, not a pair of isolated points: y is free.

Surface area and volume of revolution

Choose the geometric element before integrating

For a curve rotated about a straight axis, let $R \geq 0$ be its perpendicular distance from the axis. The **curved surface area** is

$$S = \int_C 2\pi R \, \mathrm{d}s.$$

For a graph $y = f(x)$, use $ds = \sqrt{1 + (f')^2}\,\mathrm{d}x$; for a graph $x = g(y)$, use $ds = \sqrt{1 + (g')^2}\,\mathrm{d}y$. For example,

$$\text{about the } x\text{-axis:} \quad S = 2\pi \int_a^b |f(x)| \sqrt{1 + [f'(x)]^2}\,\mathrm{d}x,$$

$$\text{about the } y\text{-axis:} \quad S = 2\pi \int_c^d |g(y)| \sqrt{1 + [g'(y)]^2}\,\mathrm{d}y.$$

These formulas assume a smooth, finitely long arc and a finite integral; add piecewise contributions if necessary. They calculate the rotated curved surface. Any end disks are separate surfaces and are added only when requested.

For **volume**, take a thin slice *perpendicular* to the axis. A disk has area πR^2; a washer with inner radius r has area $\pi(R^2 - r^2)$, where $0 \leq r \leq R$. Therefore

$$\text{horizontal axis:} \quad V = \pi \int_a^b (R(x)^2 - r(x)^2)\,\mathrm{d}x,$$

$$\text{vertical axis:} \quad V = \pi \int_c^d (R(y)^2 - r(y)^2)\,\mathrm{d}y.$$

Set $r = 0$ for disks. Radii are distances from the stated axis, which need not be a coordinate axis. Surface area uses $2\pi R\,ds$; volume uses $\pi(R^2 - r^2)$ times the slice thickness.

A square-root antiderivative used in the next example

For real u,

$$F(u) = \frac{1}{2}\left[u\sqrt{1+u^2} + \ln\left(u + \sqrt{1+u^2}\right)\right] \quad \text{satisfies} \quad F'(u) = \sqrt{1+u^2}.$$

The logarithm is defined because $\sqrt{1+u^2} > |u|$. To check the formula,

$$\frac{d}{du}\ln\left(u + \sqrt{1+u^2}\right) = \frac{1}{\sqrt{1+u^2}},$$

and hence

$$F'(u) = \frac{1}{2}\left[\sqrt{1+u^2} + \frac{u^2}{\sqrt{1+u^2}} + \frac{1}{\sqrt{1+u^2}}\right] = \sqrt{1+u^2}.$$

Example 31. Curved surfaces about the two coordinate axes

Find the curved surface areas generated by rotating:

(a) $y = \sin x$, $0 \le x \le \pi/2$, about the x-axis;

(b) $x = e^y$, $0 \le y \le \ln 2$, about the y-axis.

Solution.

(a) Radius y, length element measured using x

Since $\sin x \ge 0$, the radius is $R = \sin x$, and $ds = \sqrt{1 + \cos^2 x}\,\mathrm{d}x$. Thus

$$S_x = 2\pi \int_0^{\pi/2} \sin x \sqrt{1 + \cos^2 x}\,\mathrm{d}x.$$

Put $u = \cos x$, so $du = -\sin x\,\mathrm{d}x$. As x goes from 0 to $\pi/2$, u goes from 1 to 0. Reversing those limits removes the minus sign:

$$\begin{aligned} S_x &= -2\pi \int_1^0 \sqrt{1 + u^2}\,\mathrm{d}u = 2\pi \int_0^1 \sqrt{1 + u^2}\,\mathrm{d}u \\ &= \pi \left[u\sqrt{1 + u^2} + \ln\left(u + \sqrt{1 + u^2}\right) \right]_0^1. \end{aligned}$$

$$\boxed{S_x = \pi\left(\sqrt{2} + \ln(1 + \sqrt{2})\right)}$$

(b) Radius x, length element measured using y

Here $R = e^y > 0$, $dx/dy = e^y$, and $ds = \sqrt{1 + e^{2y}}\,\mathrm{d}y$. Consequently

$$S_y = 2\pi \int_0^{\ln 2} e^y \sqrt{1 + e^{2y}}\,\mathrm{d}y.$$

Put $u = e^y$, so $du = e^y\,\mathrm{d}y$. The limits become 1 and 2:

$$\begin{aligned} S_y &= 2\pi \int_1^2 \sqrt{1 + u^2}\,\mathrm{d}u \\ &= \pi \left[u\sqrt{1 + u^2} + \ln\left(u + \sqrt{1 + u^2}\right) \right]_1^2 \\ &= \pi \left[2\sqrt{5} - \sqrt{2} + \ln(2 + \sqrt{5}) - \ln(1 + \sqrt{2}) \right]. \end{aligned}$$

$$\boxed{S_y = \pi \left[2\sqrt{5} - \sqrt{2} + \ln\left(\frac{2 + \sqrt{5}}{1 + \sqrt{2}}\right) \right]}$$

Example 32. Washer volume about a horizontal axis

The region bounded by $x = 0$, $y = \sqrt{x}$ and $y = 2 - x$ is rotated about the x-axis. Find its volume.

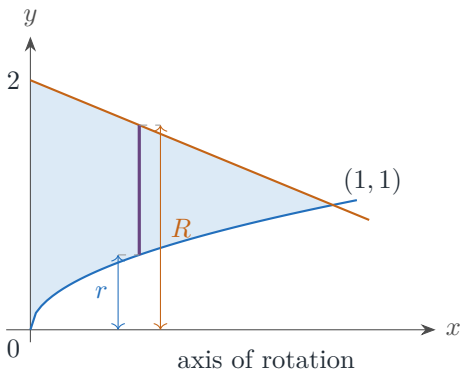
Solution.

Find the interval and the two radii

At the intersection, $\sqrt{x} = 2 - x$. Writing $u = \sqrt{x} \ge 0$,

$$u = 2 - u^2 \quad \Longrightarrow \quad (u - 1)(u + 2) = 0.$$

Only $u = 1$ is allowed, so the intersection is $(1, 1)$. The left boundary is $x = 0$; hence $0 \le x \le 1$. On this interval $2 - x \ge \sqrt{x} \ge 0$, so the outer radius is $R = 2 - x$ and the inner radius is $r = \sqrt{x}$.



Integrate the washer area

$$\begin{aligned} V &= \pi \int_0^1 \left[(2-x)^2 - (\sqrt{x})^2\right] \, dx \\ &= \pi \int_0^1 (4 - 5x + x^2) \, dx \\ &= \pi \left[4x - \frac{5}{2}x^2 + \frac{x^3}{3}\right]_0^1 = \pi \left(4 - \frac{5}{2} + \frac{1}{3}\right). \end{aligned}$$

$$\boxed{V = \frac{11\pi}{6}}$$

The missing central disk is why $\pi(\sqrt{x})^2$ is subtracted.

Example 33. Washer volume about a vertical axis

Rotate the region

$$1 \le y \le 2, \qquad 1 \le x \le \frac{2}{y}$$

about the y-axis. Find the volume.

Solution. A slice perpendicular to the vertical axis is **horizontal**, so its thickness is dy. For each $y \in [1, 2]$, the slice runs from $x = 1$ to $x = 2/y$. Both are nonnegative and $2/y \ge 1$, giving

$$R(y) = \frac{2}{y}, \qquad r(y) = 1.$$

At $y = 2$, the two boundaries meet; the washer shrinks to zero area there.

$$\begin{aligned} V &= \pi \int_1^2 \left[\left(\frac{2}{y}\right)^2 - 1^2\right] \, dy = \pi \int_1^2 (4y^{-2} - 1) \, dy \\ &= \pi \left[-\frac{4}{y} - y\right]_1^2 = \pi\big[(-2 - 2) - (-4 - 1)\big]. \end{aligned}$$

$$\boxed{V = \pi}$$

Double integrals in Cartesian coordinates

From small area cells to an integral

Divide a bounded plane region R into small area cells of areas ΔA_i. Choose a sample point (x_i^*, y_i^*) in each cell. The sum

$$\sum_i f(x_i^*, y_i^*)\,\Delta A_i$$

approximates an accumulated quantity over R. If the sums approach the same limit as the cell sizes tend to zero, independently of the subdivisions and sample points, that limit is the **double integral** $\iint_R f(x,y)\,dA$.

A useful sufficient setting is that f is continuous on a closed bounded region with a boundary made of finitely many smooth pieces. For a rectangle $R = [a,b] \times [c,d]$, Fubini's theorem then gives

$$\iint_R f\,dA = \int_a^b \int_c^d f(x,y)\,\mathrm{d}y\,\mathrm{d}x = \int_c^d \int_a^b f(x,y)\,\mathrm{d}x\,\mathrm{d}y.$$

In $dy\,dx$, integrate in y first while holding x fixed; integrate the resulting function of x next. The differential nearest the integrand belongs to the innermost integral.

General regions: bottom to top, or left to right

For continuous bounding functions and a continuous integrand:

$$R = \{a \le x \le b,\ g_1(x) \le y \le g_2(x)\} \quad\Longrightarrow\quad \iint_R f\,dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f\,\mathrm{d}y\,\mathrm{d}x,$$

$$R = \{c \le y \le d,\ h_1(y) \le x \le h_2(y)\} \quad\Longrightarrow\quad \iint_R f\,dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f\,\mathrm{d}x\,\mathrm{d}y.$$

The first is a **Type I** description: a fixed vertical slice runs from bottom g_1 to top g_2. The second is a **Type II** description: a fixed horizontal slice runs from left h_1 to right h_2. The outer limits cover the full horizontal or vertical extent. If the formula for a slice endpoint changes, split the integral at that change.

The geometric interpretations needed here are

$$\text{area}(R) = \iint_R 1\,dA, \qquad V = \iint_R (z_{\text{top}} - z_{\text{bottom}})\,dA,$$

where $z_{\text{top}} \ge z_{\text{bottom}}$ everywhere on R. An integral of a function that takes both signs is a signed accumulation; it is not automatically a geometric volume.

Example 34. Treat the other variable as a constant

Evaluate

$$I = \iint_R xe^y\,dA, \qquad R = \{0 \le x \le 2,\ 0 \le y \le \ln 3\}.$$

Solution. The integrand is continuous on the rectangle. With y first, x is a constant multiplier:

$$\begin{aligned} I &= \int_0^2 \int_0^{\ln 3} xe^y\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_0^2 x\,[e^y]_0^{\ln 3}\,\mathrm{d}x = \int_0^2 x(3-1)\,\mathrm{d}x \\ &= \Big[x^2\Big]_0^2 = 4. \end{aligned}$$

$\boxed{I = 4}$

Check. In the other order, $\int_0^{\ln 3} e^y[x^2/2]_0^2\,\mathrm{d}y = 2[e^y]_0^{\ln 3} = 4$. The factor held constant changes with the inner variable.

Example 35. The same area from two slice directions

Find the area of

$$R = \{1 \le x \le e,\ 0 \le y \le \ln x\}.$$

Then reverse the order and check the result.

Solution.

Vertical slices: the given description

For a fixed $x \in [1, e]$, the slice enters at $y = 0$ and leaves at $y = \ln x$. Thus

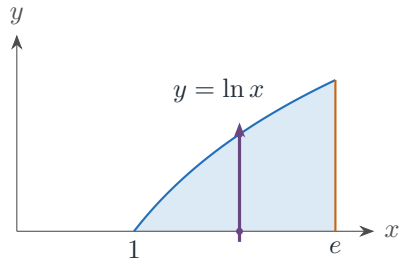
$$A = \int_1^e \int_0^{\ln x} 1 \, dy \, dx = \int_1^e \ln x \, dx.$$

Integration by parts, $\int u\, dv = uv - \int v\, du$, with $u = \ln x$, $dv = dx$, gives

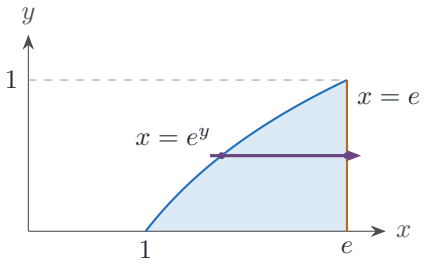
$$\int \ln x \, dx = x \ln x - \int 1 \, dx = x \ln x - x + C.$$

Consequently

$$A = [x \ln x - x]_1^e = (e - e) - (0 - 1) = 1.$$



fix x: move upward



fix y: move right

Horizontal slices: derive every new bound

The lowest point has $y = 0$; the highest has $y = \ln e = 1$. These give the new outer limits $0 \le y \le 1$. Holding such a y fixed,

$$y \le \ln x \quad \Longleftrightarrow \quad x \ge e^y.$$

The left endpoint is therefore $x = e^y$, and the right endpoint remains the vertical boundary $x = e$. Hence

$$\begin{aligned} A &= \int_0^1 \int_{e^y}^e 1 \, dx \, dy = \int_0^1 (e - e^y) \, dy \\ &= [ey - e^y]_0^1 = (e - e) - (0 - 1) = 1. \end{aligned}$$

$\boxed{A = 1}$

Example 36. Storage space between a floor and a roof

A storage space has the triangular floor plan with vertices $(0, 0)$, $(2, 0)$ and $(2, 3)$. Its lower and upper boundaries are

$$z_{\text{bottom}} = 1 + \frac{y}{3}, \qquad z_{\text{top}} = 3 + x.$$

Coordinates and heights are measured in metres. Find the available volume.

Solution.

Describe the floor plan and the vertical height

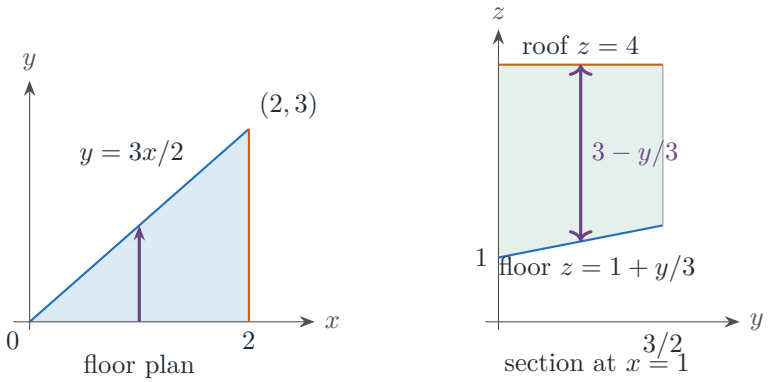
The sloping edge joins $(0, 0)$ to $(2, 3)$, so it has equation $y = 3x/2$. A vertical slice in the floor plan runs from $y = 0$ to $y = 3x/2$, while x ranges from 0 to 2:

$$R = \left\{0 \le x \le 2,\ 0 \le y \le \frac{3x}{2}\right\}.$$

The usable height is the difference between the two surfaces,

$$h(x,y) = (3+x) - \left(1 + \frac{y}{3}\right) = 2 + x - \frac{y}{3}.$$

It is positive on the whole triangle: since $y \le 3x/2$, $h \ge 2 + x/2 \ge 2$. No part of the stated floor plan needs to be removed.



Add height times small floor area

$$\begin{aligned} V &= \int_0^2 \int_0^{3x/2} \left(2 + x - \frac{y}{3}\right) \mathrm{d}y\,\mathrm{d}x \\ &= \int_0^2 \left[(2+x)y - \frac{y^2}{6}\right]_0^{3x/2} \mathrm{d}x \\ &= \int_0^2 \left[3x + \frac{3}{2}x^2 - \frac{3}{8}x^2\right] \mathrm{d}x = \int_0^2 \left(3x + \frac{9}{8}x^2\right) \mathrm{d}x \\ &= \left[\frac{3}{2}x^2 + \frac{3}{8}x^3\right]_0^2 = 6 + 3 = 9. \end{aligned}$$

$\boxed{V = 9 \text{ m}^3}$

Check. The floor-plan area is $\frac{1}{2}(2)(3) = 3\,\text{m}^2$. The calculated volume corresponds to an average height of 3 m, consistent with the stated floor and roof.

Changing the order of integration

The region stays fixed; the slices change

To reverse an iterated double integral:

1. Read its inequalities and locate every boundary and intersection.
2. Find the full range of the new outer variable from the region.
3. Hold that variable fixed and move a slice through the region. Its entry and exit points give the new inner bounds.
4. If an entry or exit boundary changes, use separate integrals on the corresponding outer-variable intervals.

Reversal changes the limits and the order of the differentials. It does not replace $f(x,y)$ by $f(y,x)$.

Example 37. A reversal that needs two pieces

For a continuous function f, reverse the order in

$$I = \int_{-1}^{2} \int_{x^2}^{4} f(x, y)\, dy\, dx.$$

Then use $f = 1$ to find the area of the region.

Solution.

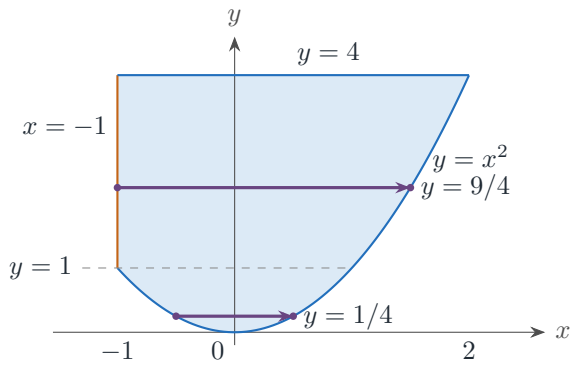
Recover the region and the full vertical range

The inequalities are $-1 \le x \le 2$ and $x^2 \le y \le 4$. The region lies above the parabola $y = x^2$, below $y = 4$, and to the right of $x = -1$. Its lowest point is the vertex $(0, 0)$, so $0 \le y \le 4$.

At a fixed nonnegative y, the parabola gives

$$x^2 \le y \quad \Longleftrightarrow \quad -\sqrt{y} \le x \le \sqrt{y}.$$

We must also retain $-1 \le x \le 2$. The right endpoint is always $\sqrt{y}$, because $\sqrt{y} \le 2$ on $[0, 4]$. The left endpoint is whichever is farther right: $-\sqrt{y}$ or -1.



Locate the switch and test one slice on each side

The two possible left boundaries meet when $-\sqrt{y} = -1$, that is, at $y = 1$.

- If $0 \le y \le 1$, then $-\sqrt{y} \ge -1$, so the slice runs from $-\sqrt{y}$ to $\sqrt{y}$. For example, at $y = 1/4$ it is $[-1/2, 1/2]$.
- If $1 \le y \le 4$, then $-\sqrt{y} \le -1$, so the left wall cuts off part of that slice. It runs from -1 to $\sqrt{y}$. At $y = 9/4$ it is $[-1, 3/2]$.

Therefore

$$\boxed{I = \int_{0}^{1} \int_{-\sqrt{y}}^{\sqrt{y}} f(x, y)\, dx\, dy + \int_{1}^{4} \int_{-1}^{\sqrt{y}} f(x, y)\, dx\, dy.}$$

Set the integrand equal to 1 for area

$$\begin{aligned} A &= \int_{0}^{1} 2\sqrt{y}\, dy + \int_{1}^{4} (\sqrt{y} + 1)\, dy \\ &= \left[\frac{4}{3} y^{3/2}\right]_{0}^{1} + \left[\frac{2}{3} y^{3/2} + y\right]_{1}^{4} \\ &= \frac{4}{3} + \left(\frac{16}{3} + 4 - \frac{2}{3} - 1\right) = 9. \end{aligned}$$

$\boxed{A = 9}$

Check. In the original order, $\int_{-1}^{2} (4 - x^2)\, dx = [4x - x^3/3]_{-1}^{2} = 9$. One integral in the original order becomes two after reversal.

Example 38. Reversal makes the inner integration elementary

Evaluate

$$I = \int_0^{\pi/2} \int_y^{\pi/2} \frac{\sin x}{x}\, \mathrm{d}x\, \mathrm{d}y.$$

At $x = 0$, use the continuous extension $\sin x/x = 1$.

Solution. The function $\sin x/x$ has no elementary antiderivative in x. Instead, read the inequalities:

$$0 \le y \le \frac{\pi}{2}, \qquad y \le x \le \frac{\pi}{2}.$$

They describe the triangle below $y = x$, above $y = 0$, and to the left of $x = \pi/2$. Its full horizontal range is $0 \le x \le \pi/2$. For each fixed x in that interval, a vertical slice starts at $y = 0$ and ends at $y = x$. Hence

$$I = \int_0^{\pi/2} \int_0^{x} \frac{\sin x}{x}\, \mathrm{d}y\, \mathrm{d}x.$$

The inner variable is now y, so $\sin x/x$ is constant during that integration. For $x > 0$,

$$\int_0^{x} \frac{\sin x}{x}\, \mathrm{d}y = \frac{\sin x}{x}[y]_0^x = \frac{\sin x}{x}\, x = \sin x.$$

The same result extends continuously to $x = 0$. Therefore

$$I = \int_0^{\pi/2} \sin x\, \mathrm{d}x = [-\cos x]_0^{\pi/2} = 0 - (-1) = 1.$$

$\boxed{I = 1}$

Why the change helps

The new slice has length x. Integrating in y multiplies the unchanged function $\sin x/x$ by this length, cancelling the denominator. The continuity convention at the origin makes the integrand continuous on the closed triangle; it does not add any extra area.

Polar double integrals and conversion in both directions

Change the coordinates, the region and the area element

Standard polar coordinates are measured from the original origin:

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad x^2 + y^2 = r^2, \qquad r \ge 0.$$

A small polar cell has radial width dr and approximate transverse width $r\, d\theta$, so its area is $r\, dr\, d\theta$. The same factor follows from the coordinate derivatives:

$$\left|\frac{\partial(x,y)}{\partial(r,\theta)}\right| = \left\|\begin{matrix}\cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta\end{matrix}\right\| = r, \qquad dA = r\, \mathrm{d}r\, \mathrm{d}\theta.$$

If R is described once by $\alpha \le \theta \le \beta$, $a(\theta) \le r \le b(\theta)$, with $0 \le a \le b$, then for a continuous integrand

$$\iint_R f(x,y)\, dA = \int_\alpha^\beta \int_{a(\theta)}^{b(\theta)} f(r\cos\theta, r\sin\theta)\, r\, \mathrm{d}r\, \mathrm{d}\theta.$$

Choose the angular interval from the rays that reach the region. On each fixed ray, choose the radial limits from its entry and exit points. Repeated descriptions on boundary rays or at the origin do not affect area; avoid covering a positive-area part of the region twice.

For the reverse conversion, identify which factor in the polar integral is the area factor r. The remaining expression represents $f(x,y)$. Then describe the same region using Cartesian slices.

Example 39. A volume that becomes simpler in polar coordinates

Find the volume below

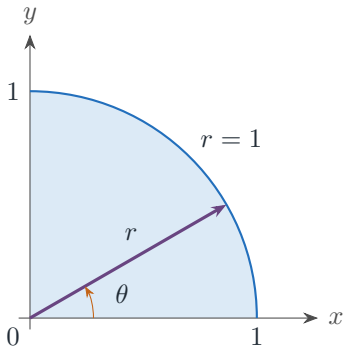
$$z = \frac{1}{1 + x^2 + y^2}$$

and above the first-quadrant quarter disk $R = \{x \geq 0,\ y \geq 0,\ x^2 + y^2 \leq 1\}$ in the xy-plane.

Solution. The lower surface is $z = 0$, and the upper surface is positive everywhere. Its height is therefore $1/(1 + x^2 + y^2)$. In Cartesian coordinates,

$$V = \int_0^1 \int_0^{\sqrt{1-x^2}} \frac{1}{1 + x^2 + y^2}\, \mathrm{d}y\, \mathrm{d}x.$$

Both the circular boundary and the expression $x^2 + y^2$ suggest polar coordinates.



Derive the new bounds and the new integrand

The first quadrant gives $0 \leq \theta \leq \pi/2$. Each ray starts at the origin and leaves the disk at $r = 1$, so $0 \leq r \leq 1$. The height becomes $1/(1 + r^2)$, while the area element contributes an additional factor r:

$$V = \int_0^{\pi/2} \int_0^1 \frac{r}{1 + r^2}\, \mathrm{d}r\, \mathrm{d}\theta.$$

In the radial integral, put $u = 1 + r^2$, $du = 2r\, \mathrm{d}r$. Its limits change from $r = 0, 1$ to $u = 1, 2$:

$$\int_0^1 \frac{r}{1 + r^2}\, \mathrm{d}r = \frac{1}{2} \int_1^2 \frac{1}{u}\, \mathrm{d}u = \frac{1}{2}[\ln u]_1^2 = \frac{1}{2} \ln 2.$$

Thus

$$V = \int_0^{\pi/2} \frac{1}{2} \ln 2\, \mathrm{d}\theta = \frac{1}{2} \ln 2\, [\theta]_0^{\pi/2}.$$

$$\boxed{V = \frac{\pi}{4} \ln 2 \quad \text{cubic units}}$$

Check. The height lies between $1/2$ and 1, and the base area is $\pi/4$. The volume therefore lies between $\pi/8$ and $\pi/4$, as the answer does.

Example 40. A circle whose centre is not the polar origin

Use standard polar coordinates to find the area of

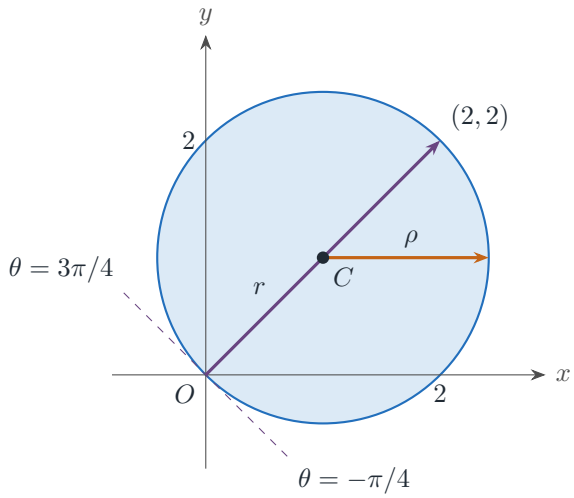
$$R = \{x^2 + y^2 \leq 2x + 2y\}.$$

Check the result using polar coordinates measured from the circle's centre.

Solution. Completing squares gives

$$(x-1)^2 + (y-1)^2 \leq 2.$$

The circle has centre $C = (1,1)$ and radius $\sqrt{2}$. Its boundary passes through the original origin. The radius $\sqrt{2}$ is measured from C, whereas standard r is measured from $O = (0,0)$.



Standard polar coordinates: first determine the rays

Substitute $x = r\cos\theta$, $y = r\sin\theta$:

$$r^2 \leq 2r(\cos\theta + \sin\theta).$$

For $r > 0$, division by r gives $r \leq 2(\cos\theta + \sin\theta)$. Such a positive radial interval exists when $\cos\theta + \sin\theta > 0$. Since

$$\cos\theta + \sin\theta = \sqrt{2}\cos(\theta - \pi/4),$$

one complete choice of rays, including the two boundary rays, is

$$-\frac{\pi}{4} \leq \theta \leq \frac{3\pi}{4}, \qquad 0 \leq r \leq 2(\cos\theta + \sin\theta).$$

The origin, excluded temporarily when dividing by r, is included by the lower limit $r = 0$.

Integrate the area element

$$\begin{aligned} A &= \int_{-\pi/4}^{3\pi/4} \int_0^{2(\cos\theta+\sin\theta)} r \, \mathrm{d}r \, \mathrm{d}\theta \\ &= 2\int_{-\pi/4}^{3\pi/4} (\cos\theta + \sin\theta)^2 \, \mathrm{d}\theta \\ &= 2\int_{-\pi/4}^{3\pi/4} (1 + \sin 2\theta) \, \mathrm{d}\theta \\ &= [2\theta - \cos 2\theta]_{-\pi/4}^{3\pi/4} = \frac{3\pi}{2} - \left(-\frac{\pi}{2}\right) = 2\pi. \end{aligned}$$

$\boxed{A = 2\pi}$

Coordinates measured from the centre: state the new map

With

$$x = 1 + \rho\cos\phi, \qquad y = 1 + \rho\sin\phi,$$

the same disk has $0 \le \rho \le \sqrt{2}$, $0 \le \phi \le 2\pi$, and $dA = \rho\, d\rho\, d\phi$. Hence

$$A = \int_0^{2\pi}\int_0^{\sqrt{2}} \rho\,\mathrm{d}\rho\,\mathrm{d}\phi = \int_0^{2\pi}\left[\frac{\rho^2}{2}\right]_0^{\sqrt{2}}\mathrm{d}\phi = 2\pi.$$

The constant radial limit $\sqrt{2}$ belongs to the translated map, not to standard polar coordinates about O.

Example 41. Recognising a Cartesian region from polar limits

Let $\alpha = \tan^{-1}(1/2)$. Convert the following sum to a Cartesian double integral and evaluate it:

$$I = \int_0^{\alpha}\int_0^{2\sec\theta} r(1 + r\sin\theta)\,\mathrm{d}r\,\mathrm{d}\theta + \int_{\alpha}^{\pi/2}\int_0^{\csc\theta} r(1 + r\sin\theta)\,\mathrm{d}r\,\mathrm{d}\theta.$$

Solution.

Translate each exit boundary

For the first angular interval, $\cos\theta > 0$, so

$$r \le 2\sec\theta \iff r\cos\theta \le 2 \iff x \le 2.$$

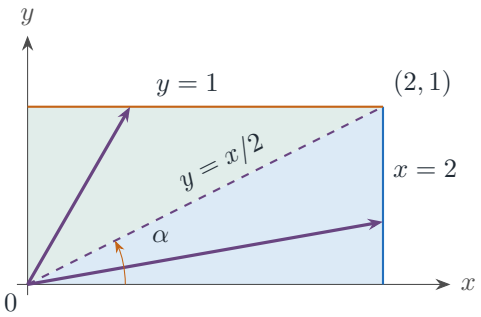
The exit boundary is the vertical line $x = 2$, not a circle. For the second interval, $\sin\theta > 0$, so

$$r \le \csc\theta \iff r\sin\theta \le 1 \iff y \le 1.$$

This exit boundary is the horizontal line $y = 1$.

The separating ray has slope $\tan\alpha = 1/2$, so it is $y = x/2$. It passes through $(2, 1)$. Rays below it reach $x = 2$ first; rays above it reach $y = 1$ first. Together the two pieces cover the rectangle

$$R = \{0 \le x \le 2,\ 0 \le y \le 1\}.$$



Separate the function from the area factor

Since $r\sin\theta = y$,

$$r(1 + r\sin\theta)\,\mathrm{d}r\,\mathrm{d}\theta = (1 + y)\,dA.$$

The Cartesian integrand is $1 + y$; the leading r has become part of dA. Therefore

$$I = \int_0^2\int_0^1 (1 + y)\,\mathrm{d}y\,\mathrm{d}x = \int_0^2\left[y + \frac{y^2}{2}\right]_0^1\mathrm{d}x$$
$$= \int_0^2 \frac{3}{2}\,\mathrm{d}x = 3.$$

$\boxed{I = 3}$

Cartesian triple integrals: building a volume integral

From small boxes to dV

A double integral adds contributions over small *area* cells. For a solid region D, partition a box containing D into small rectangular boxes and retain the boxes lying inside D. Under the regularity assumptions below, the total volume of the uncovered boundary pieces tends to zero as the boxes shrink. A box with side lengths $\Delta x_i, \Delta y_i, \Delta z_i$ has volume

$$\Delta V_i = \Delta x_i \, \Delta y_i \, \Delta z_i.$$

Choose a point (x_i^*, y_i^*, z_i^*) in each box. The corresponding sum is

$$\sum_i f(x_i^*, y_i^*, z_i^*) \, \Delta V_i.$$

As all box dimensions tend to zero, the common limit, when it exists independently of the subdivisions and sample points, is

$$\iiint_D f(x, y, z) \, dV.$$

A sufficient setting is a continuous function on a closed bounded solid whose boundary consists of finitely many smooth surface pieces. Cartesian volume elements are products of the three coordinate widths, such as $dV = dz\, dy\, dx$; there is no polar factor r.
If $f = 1$, every small-box contribution is its volume. Thus

$$\boxed{\text{volume}(D) = \iiint_D 1 \, dV.}$$

For a general f, the triple integral accumulates f over the solid; its value need not equal the solid's volume.

Fubini's theorem on a rectangular box

Let $D = [a, b] \times [c, d] \times [e, h]$, and let f be continuous on D. Then

$$\iiint_D f \, dV = \int_a^b \int_c^d \int_e^h f(x, y, z) \, \mathrm{d}z \, \mathrm{d}y \, \mathrm{d}x.$$

Read this from the inside outward:

1. Hold x, y fixed and integrate in z from e to h.
2. Hold x fixed and integrate the resulting function in y from c to d.
3. Integrate the resulting function of x from a to b.

On a box, any order is possible if each coordinate keeps its own range.

Example 42. A box: volume and a different accumulated quantity

For

$$D = \{1 \le x \le 2,\ -1 \le y \le 1,\ 0 \le z \le 1\},$$

calculate both volume(D) and $\iiint_D (1 + xz)\, dV$.

Solution.

Volume uses the integrand 1

$$V = \int_1^2 \int_{-1}^1 \int_0^1 1 \,\mathrm{d}z \,\mathrm{d}y \,\mathrm{d}x = \int_1^2 \int_{-1}^1 [z]_0^1 \,\mathrm{d}y \,\mathrm{d}x$$
$$= \int_1^2 [y]_{-1}^1 \,\mathrm{d}x = \int_1^2 2 \,\mathrm{d}x = 2.$$

This agrees with the side-length product $(2-1)(1-(-1))(1-0) = 2$.

Integrate $1 + xz$ in the same order

During the z-integration, x is constant:

$$\int_0^1 (1 + xz) \,\mathrm{d}z = \left[z + \frac{xz^2}{2} \right]_0^1 = 1 + \frac{x}{2}.$$

This result has no y in it, so the y-integration multiplies it by the interval length 2:

$$\int_{-1}^1 \left(1 + \frac{x}{2}\right) \mathrm{d}y = \left[\left(1 + \frac{x}{2}\right) y \right]_{-1}^1 = 2 + x.$$

Finally,

$$\iiint_D (1 + xz)\, dV = \int_1^2 (2 + x) \,\mathrm{d}x = \left[2x + \frac{x^2}{2} \right]_1^2 = 6 - \frac{5}{2} = \frac{7}{2}.$$

$$\text{volume}(D) = 2, \qquad \iiint_D (1 + xz)\, dV = \frac{7}{2}$$

A variable-bound solid: projection, floor and roof

The **projection** of a solid onto the xy-plane is its footprint: the set of (x, y) for which at least one point of the solid exists above or below that location. Suppose the footprint is

$$R = \{a \le x \le b,\ p(x) \le y \le q(x)\},$$

and, at every point of R, the solid extends from a lower surface $z = s(x, y)$ to an upper surface $z = t(x, y)$, with $s \le t$. Then

$$\iiint_D f\, dV = \int_a^b \int_{p(x)}^{q(x)} \int_{s(x,y)}^{t(x,y)} f(x, y, z) \,\mathrm{d}z \,\mathrm{d}y \,\mathrm{d}x.$$

At the innermost step, x, y are fixed, so the z-bounds may depend on them. At the middle step, the y-bounds may depend on x. The outer x-limits are constants. No bound may use its own integration variable or a variable that has already been integrated out.
The floor and roof must each be the endpoint of a single vertical segment. If a vertical line meets separated pieces of a more complicated solid, split those pieces before applying this formula.

Example 43. A solid with a curved footprint and a raised floor

A solid has footprint bounded by $x = 0$, $y = 0$ and $y = 1 - x^2$ in the first quadrant. Its lower surface is $z = y$, and its upper surface is $z = 2 - x^2$. Find its volume using a Cartesian triple integral, and explain the connection with a double integral of height.

Solution.

1. Derive the three ranges

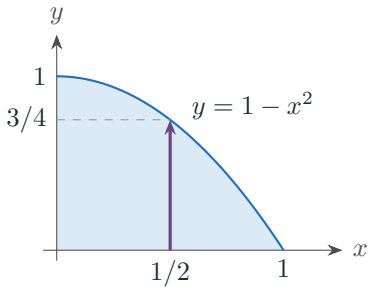
The curved edge meets $y = 0$ where $1 - x^2 = 0$, giving $x = \pm 1$. The first-quadrant condition selects $x = 1$. Thus the full horizontal range is $0 \le x \le 1$.

For each such x, the footprint runs from the lower edge $y = 0$ to the curved edge $y = 1 - x^2$. For each fixed point (x, y) in that footprint, the vertical segment runs from $z = y$ to $z = 2 - x^2$. Together these give

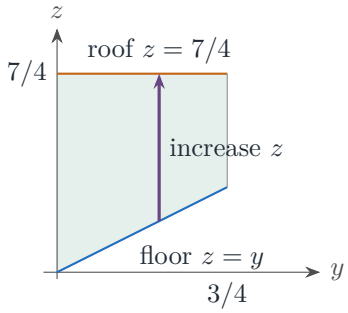
$$\boxed{0 \le x \le 1, \qquad 0 \le y \le 1 - x^2, \qquad y \le z \le 2 - x^2.}$$

The upper surface stays above the lower one, because

$$(2 - x^2) - y \ge (2 - x^2) - (1 - x^2) = 1.$$



projection onto the xy-plane



section at $x = 1/2$

2. Integrate in z: each vertical segment contributes its height

$$\begin{aligned} V &= \int_0^1 \int_0^{1-x^2} \int_y^{2-x^2} 1 \, dz \, dy \, dx \\ &= \int_0^1 \int_0^{1-x^2} [z]_y^{2-x^2} \, dy \, dx = \int_0^1 \int_0^{1-x^2} (2 - x^2 - y) \, dy \, dx. \end{aligned}$$

3. Integrate in y, keeping x fixed

$$\begin{aligned} V &= \int_0^1 \left[(2 - x^2)y - \frac{y^2}{2} \right]_0^{1-x^2} dx \\ &= \int_0^1 \left[(2 - x^2)(1 - x^2) - \frac{1}{2}(1 - x^2)^2 \right] dx. \end{aligned}$$

Expand both terms before combining them:

$$\begin{aligned} (2 - x^2)(1 - x^2) &= 2 - 3x^2 + x^4, \\ \tfrac{1}{2}(1 - x^2)^2 &= \tfrac{1}{2} - x^2 + \tfrac{1}{2}x^4. \end{aligned}$$

Their difference is $\frac{3}{2} - 2x^2 + \frac{1}{2}x^4$.

4. Finish the outer integral

$$\begin{aligned} V &= \int_0^1 \left(\frac{3}{2} - 2x^2 + \frac{1}{2}x^4\right) \mathrm{d}x \\ &= \left[\frac{3}{2}x - \frac{2}{3}x^3 + \frac{1}{10}x^5\right]_0^1 = \frac{3}{2} - \frac{2}{3} + \frac{1}{10} = \frac{14}{15}. \end{aligned}$$

$$\boxed{V = \frac{14}{15} \quad \text{cubic units}}$$

Why a double integral of height gives the same volume

For each fixed (x, y), integrating 1 from the floor to the roof gives their height difference. In general,

$$\iiint_D 1 \, dV = \iint_R \left(\int_{z_{\text{bottom}}}^{z_{\text{top}}} 1 \, \mathrm{d}z\right) dA = \iint_R (z_{\text{top}} - z_{\text{bottom}}) \, dA.$$

Here that height is $2 - x^2 - y$, exactly the double-integral integrand obtained in Step 2. The triple integral starts with 1; its inner integration creates the height. Inserting the height again as the triple-integral integrand would multiply by the height twice.

Check. The footprint area is $\int_0^1 (1-x^2) \, \mathrm{d}x = 2/3$, and the height lies between 1 and 2. Thus $2/3 \le V \le 4/3$, consistent with $14/15$.